Blockchain Fee Policies: Quantity- vs Price-Control Design across Protocols

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Abstract

This paper develops an economic theory to explain the optimal choice of blockchain transaction fee policies, contrasting quantity controls (as in Bitcoin) and price controls (as in Ethereum). I model the blockchain as a decentralized platform where validators, possessing temporary monopoly power and facing uncertain operational costs, choose transactions under varying user demand. Price controls outperform quantity controls when demand volatility is significant, the correlation between marginal costs and demand is low, and validators hold substantial bargaining power—conditions aligning closely with Ethereum's current structure following its proof-of-stake transition. Conversely, quantity controls are optimal when marginal costs positively correlate with demand and validators' bargaining power is limited, conditions characteristic of Bitcoin's proof-of-work model. To a first-order approximation, I characterize the dynamics of Ethereum's fee policies: deviations from the block-size target decay with persistence such that adjustment faster than the inverse elasticity buys not faster convergence but oscillations and excess fee volatility. Using recent transaction data, I illustrate that Ethereum's base fee adjusts more rapidly than optimal. Additionally, I derive bounds for block size targets to mitigate maximal extractable value (MEV) exploitation. These insights offer guidance for improving blockchain governance and optimizing fee policy design.

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1 Introduction

Transaction costs play an essential role in the evolving areas of blockchain technology and decentralized finance. They manage the allocation of space on a blockchain (block space) which is a limited resource due to the economic and scaling limits of blockchain systems (Buterin, 2021; Budish, 2025). However, these transaction costs can induce illiquidity and create trade execution risk. As cryptocurrencies integrate into mainstream financial systems, exemplified by the approval of Bitcoin and Ethereum exchange-traded funds (ETFs) (SEC, 2024), inefficient fee policies can delay settlement of underlying crypto-assets and potentially affect financial stability. This paper provides a theory of how and why fee policies differ across blockchain platforms and what these fee policies should look like.

From a regulatory perspective, understanding the mechanisms behind how transaction services are priced and prioritized is crucial for developing effective oversight policies for blockchain platforms. Furthermore, from a blockchain design perspective, transaction costs directly affect economic efficiency and ultimately impact user adoption.

The Bitcoin and Ethereum blockchains are two leading examples that reflect opposite extremes in the choice of fee policies. For Bitcoin fees, a maximum quantity is set by the protocol (maximum block size), and users and transaction service providers independently choose fees. I call this the quantity-setting or quantity controls regime. On the other hand, Ethereum sets a minimum price while the block size adjusts in response to demand. I call this the price-setting or price controls regime. Most blockchains follow a variation of either the Bitcoin or the Ethereum fee policies, and some set constant fees by default or subsidize user fees until their network matures and faces congestion issues. In addition, there is an active debate on the future of the Bitcoin blockchain. As the payoff for transaction service providers programmatically decreases every four years, the role of fees becomes increasingly important for Bitcoin.

In this paper, I model a blockchain (similar to that underlying Bitcoin or Ethereum) as a distributed computing network where users submit transactions for inclusion by validators, the transaction service providers. Transactions, representing data that modify the network's state (such as account balance transfers), are submitted by users with a bid to a publicly observable pool (known as the mempool) that indicates their willingness to pay for transaction processing. Using their limited resources, validators select a subset of transactions from the mempool to form a block. Each block, comprising an ordered sequence of transactions and a reference to the previous block, can be appended to the blockchain in every period. However, technological limitations impose a maximum block size, constraining block space supply.

On the supply side of transaction processing services, validators (also known as miners in proof-of-work systems) are responsible for processing transactions and adding them to the blockchain. While there are multiple validators in the network, each validator enjoys temporary monopoly power when they are selected to propose the next block due to the consensus mechanism (e.g., proof of work or proof of stake).[1] This grants them discretion over which transactions to include and how to prioritize them. Therefore, in our model, we focus on a representative validator with monopolistic characteristics during their turn to validate a block.[2]

Producing a block is costly, and part of this cost depends on the block's contents. Including an additional transaction—equivalently, producing a larger block—consumes computation to verify signatures and execute code, bandwidth to propagate the block, and storage to update the network's state; under proof of work, a larger block also propagates more slowly and so faces a higher risk of being discarded ("orphaned") by the network. The costs of participating in consensus itself—hashing in proof of work, staking in proof of stake—are, by contrast, fixed with respect to block contents: they determine the probability of being selected to produce a block, and hence expected profits, but not the cost of including one more transaction.[3] Accordingly, I model the validator's cost of producing a block as increasing in the block size, and I set fixed participation costs aside from the block-composition margin: they matter through the validator's profits, which the analysis requires to remain high enough for validators to participate (Section 3.1 formalizes this cost structure).

The marginal cost of block space—the additional cost of a larger block or of one more transaction—varies over time, and the validator faces uncertainty about it. Electricity is a natural source of this variation: the marginal cost of producing power moves with the season and the hour of the day, and wholesale prices inherit this variation (Borenstein and Holland, 2005; Joskow and Wolfram, 2012); accordingly, the electricity costs of Bitcoin mining are large and time-varying (Delgado-Mohatar et al., 2019), and the Bitcoin hash rate responds to energy commodity prices (Rehman and Kang, 2021). The marginal cost can, moreover, be correlated with demand: under proof of work, the expected loss from an orphaned block is proportional to the value at stake (the block reward plus fees) which is precisely high when demand is high. In a proof-of-stake protocol such as the updated Ethereum, by contrast, the marginal cost of block space is essentially constant. The level of the marginal cost, its uncertainty, and its covariance with demand are exactly the objects that determine the

[1]See Saleh (2021) for an economic analysis of proof-of-stake.

[2]We interchangeably use the plural term 'validators' for the group of validators as a whole and the singular 'validator' when referring to the interim monopolist validator chosen to process transactions.

[3]In the case of monopolistic miners (such as a mining pool), Cong et al. (2021) show that the risk of being selected as a validator can be diversified.

choice of fee policy in my analysis.

On the demand side, atomistic users arrive at random and submit their transactions along with their willingness to pay for these transactions to be included in the next block. In the baseline analysis, users reveal their maximum willingness-to-pay (their true valuations) for transaction inclusion. Rather than simply assuming truthful bidding, I characterize in Section 3.2 the conditions under which it is incentive-compatible under each regime. Under price controls, truthful bidding is exactly incentive-compatible: the base-fee mechanism of EIP-1559 is dominant-strategy incentive-compatible for users and incentive-compatible for myopic validators (Roughgarden, 2020, 2021; Chung and Shi, 2023). Under quantity controls, where fees are determined by a pay-as-bid auction as in Bitcoin, truthful bidding fails for inframarginal users. I show, however, that this failure does not overturn the paper's welfare comparison: because users are atomistic and the mempool is public, equilibrium bids settle at the market-clearing price whenever block space is scarce, so that the allocation of block space and validator revenue coincide with their values under the market-clearing benchmark of the demand curve below, in which included users pay the clearing price (Proposition 1). This process forms the microfoundations of an aggregate user demand curve, with a demand shifter that captures low and high-demand situations.

In the face of these uncertainties, the protocol must commit to either a base fee—an ex-ante price control—or a block size limit—an ex-ante quantity control.[4] This choice highlights a disagreement between the blockchain designers' objective and the profit-maximization objective of the validator with significant interim monopoly power after they are chosen to validate a block. Indeed, a protocol designer who prefers full capacity utilization or has an exogenous technological block size target (that internalizes social benefits and costs) will choose the quantity-setting regime as it insulates the realized block size from demand and cost fluctuations whenever demand suffices to fill it. On the other hand, a monopolist validator would choose either the price-setting or the quantity-setting regimes, depending on the relative degree of uncertainty in demand and marginal costs.

I model the resolution of the conflict between the blockchain designer's and monopolistic validator's preferences through Nash bargaining over the protocol profits. This bargaining formulation is not a departure from standard mechanism design: I show that it is equivalent to a problem in which the protocol maximizes its own objective subject to a participation constraint guaranteeing validators a minimum expected profit, with the bargaining weight indexing the guaranteed rents (Lemma 2). I show that, in this context, the choice of instru-

[4]Since the fee policies are rule-based and encoded in the blockchain, the protocol must commit to a supply schedule before any uncertainty realizations. We analyze the two extremes of price-setting and quantity-setting supply schedules, motivated by the Bitcoin and Ethereum cases, and as a first step in the analysis of blockchain fee policies.

ments is more nuanced than Weitzman's (1974) "prices vs. quantities" insight: namely, that price controls prove more effective when demand uncertainty is high and quantity controls more effective when uncertainty in marginal costs is high. In general, the key determinants of the advantage of price-setting in blockchains are the validator's bargaining power, the elasticity of demand, the validator's uncertainty about demand, and the covariance of demand and marginal costs.

First, demand uncertainty favors price controls when the validator has high bargaining power, as block size adjustments provide the flexibility necessary to accommodate demand fluctuations. Second, a positive correlation between marginal costs and demand disfavors price controls. In this case, quantity adjustments would produce larger blocks when marginal costs are high, thereby decreasing efficiency. Third, a higher price elasticity of demand—the proportional change in aggregate block space demand in response to a proportional change in price, evaluated at the valuation of the marginal user seeking inclusion in the next block—further amplifies the relative advantage of price controls over quantity controls and makes the choice over fee policies even more important. However, quantity controls become more effective if the monopolist validator has low bargaining power. Last, without uncertainty, the blockchain designer and validator remain indifferent between price and quantity controls.

It is worth being precise about what in this analysis is blockchain-specific, beyond relabeling the classical prices-versus-quantities logic. Three features distinguish the blockchain problem from Weitzman's (1974) regulation problem. First, the regulated party is not a price-taking firm but an *interim monopolist*: the consensus mechanism grants the selected validator full discretion over block contents, so any fee policy must be evaluated against the strategic response of a monopolist who can undo it ex-post—for instance, by leaving blocks partially empty or by stuffing them with her own transactions. Second, the "regulator" is not a government with coercive power but a protocol that can only commit to rules encoded in software ex-ante and whose continued operation requires the voluntary participation of the validators themselves; this is why the designer's and validator's objectives enter jointly through a bargaining problem, and why the bargaining weight β—which I map to observable governance differences between Bitcoin and Ethereum—is a key comparative static. Third, the instruments themselves are constrained by the technology: price controls must be quoted in a volatile native currency (Section 3.5), quantity controls interact with maximal extractable value (Section 4), and the adjustment of either instrument can only respond to on-chain observables such as realized block sizes. Each of these features matters: the optimal instrument depends on the validator's bargaining power, the optimal adjustment speed is pinned down by an elasticity that must be measured on-chain, and currency volatility shifts the comparison toward quantity controls.

This paper's question is also distinct from, and complementary to, the literature on existing transaction fee mechanisms (TFMs), which takes a mechanism as given and characterizes its incentive properties—for whom is truthful bidding optimal, and when will validators follow the rules. One strand studies Ethereum's TFM in this way (Roughgarden, 2020, 2021; Chung and Shi, 2023); another takes Bitcoin's TFM as given and studies its incentive and security properties (Sompolinsky and Zohar, 2015; Huberman et al., 2021; Lehar and Parlour, 2020). I ask the prior design question: *which* family of fee policy—a price instrument or a quantity instrument—should the protocol commit to in the first place, given uncertainty in demand and costs and the division of surplus between the protocol and its validators, and *how* should the chosen instrument adjust in a changing environment. The incentive-compatibility results of that literature serve as an input to my analysis (Section 3.2), while my welfare comparison, optimal adjustment-speed formula, and MEV-robust target bounds have no counterpart in it.

This result helps us understand the differences in the policies determining fees and block space in Bitcoin and Ethereum. For both blockchains, the price elasticity of demand is high, and demand uncertainty is significant.

Ethereum can be viewed through the lens of our model as an instance where validators have high bargaining power (or are considered influential by the protocol designers).[5] In addition, since its proof-of-stake upgrade, the marginal cost of additional block space for Ethereum validators is virtually constant. In this case, my result points to price-setting as the most favorable policy for Ethereum. This helps explain the recent adoption of the Ethereum Improvement Proposal 1559's (EIP-1559's) price-setting policy for the Ethereum blockchain. Furthermore, Ethereum's fee policies are found to perform better after Ethereum's proof-of-stake upgrade, consistent with the correlation between validators' marginal costs and demand being lower than miners' in a proof-of-work protocol.

[5]While the Ethereum Foundation (EF) maintains considerable authority over protocol upgrades, the validators' economic leverage in day-to-day block proposal can also be significant. In particular, validators may capture substantial amounts of fees and Maximum Extractable Value (MEV), and hence, they can indirectly influence or resist certain protocol-level fee adjustments if these undermine their profitability. Nonetheless, the Ethereum governance process—rooted in Ethereum Improvement Protocols (EIPs) and developer-led forks—often appears to grant the EF a strong position. Fracassi et al. (2024) find that 10 individuals are responsible for proposing 68% of all implemented Core EIPs, some of which are connected to Consensys, a private company, and other EF affiliates that have advised private companies. Recent comparisons of the Ethereum blockchain to its competitor Solana also led to a push to increase its block size limit (`https://pumpthegas.org/`). Since the first version of this paper, that push has succeeded: validators raised the gas limit from 30 million—its level since the London hard fork—to 36 million in February 2025, 45 million in July 2025, and 60 million in November 2025, a doubling codified as the client default by the Fusaka upgrade in December 2025 (Nijkerk, 2025; Ethereum Foundation, 2025). Because EIP-1559 sets the target at half the limit, the block size target doubled correspondingly. These limit increases require no hard fork—they activate once a majority of validators signal them—and they illustrate the validators' collective leverage over supply parameters that the bargaining weight β captures.

For its part, Bitcoin is distinguished as the first blockchain with an anonymous founder, and its core developers strongly emphasize decentralization and censorship resistance. Most proposals to change Bitcoin's fees and block size policies have failed. Bitcoin can, therefore, be viewed through the lens of our model as an instance where validators (here miners) have low bargaining power (or are not attributed enough importance by protocol designers). In addition, under the proof-of-work protocol, the marginal cost of miners is largely positively correlated with demand—the relevant cost being, as discussed above, the cost of marginal block space conditional on having been selected, not the cost of the hashing race. The correlation operates primarily through orphan risk, whose expected loss is proportional to the value at stake and hence high when demand is high, and more modestly through the per-transaction component of electricity costs, since power prices co-move with the cryptocurrency market cycles that drive transaction demand (Rehman and Kang, 2021; Delgado-Mohatar et al., 2019). Our result then states that, in this case, price-setting is less effective than in the case of Ethereum.

Last, it is essential to consider that users (and validators) might value block space (and marginal costs) in dollars or real terms rather than in the native currency of the blockchain. To accommodate this, I expand the model to introduce uncertainty in the cryptocurrency price in US dollars or real terms. Notably, price controls are restricted to be expressed in units of the native currency. I show that volatility in the cryptocurrency price reduces the advantage of price controls over quantity controls.

Building on these insights, I briefly discuss the implications of my results for fee policies on Ethereum, the most widely utilized public blockchain, whose fee policy is a blueprint for many blockchains that follow the price-setting regime. I show that the rate at which Ethereum's fee policies readjust prices should be tightly linked to the price elasticity of demand to include a transaction in the next block. Using a random sample of Ethereum transaction data, I apply my framework to evaluate recent changes to Ethereum's fee policies. Every estimate—offered as an illustrative benchmark, given the limitations of observed fees as a proxy for valuations—places the rate at which Ethereum's fees are changed above the implied optimum. Beyond the level of the rate, I characterize the dynamics it induces. The logic is that of a feedback rule. Each block, the protocol moves the base fee in proportion to the previous block's deviation from the target, and demand responds to the new fee in proportion to its elasticity; the product of the adjustment rate and the demand elasticity therefore measures how much of a deviation one round of adjustment undoes. Block-size deviations follow an autoregressive process whose persistence is one minus this product. When the product is below one, the fee under-reacts and deviations fade monotonically; at exactly one, a single adjustment returns demand to the target; beyond one, the fee overcorrects, so that a positive deviation is followed by a negative one and fees and block sizes oscillate around the target; beyond two, each over-correction exceeds the previous deviation and the dynamics become locally unstable. The same threshold is also the welfare optimum: an adjustment rate equal to the inverse demand elasticity uniquely minimizes the stationary volatility of both fees and block sizes. This provides a theoretical account of the stability thresholds that prior work documented in simulations of EIP-1559 (Leonardos et al., 2023), and it implies that, at my elasticity estimates, Ethereum's current 12.5% adjustment rate places the fee dynamics in the oscillatory region.

In addition, I characterize the optimal block size target (a quantity control) for a monopolistic validator and offer tight bounds on its size relative to the block size limit. There are widespread concerns regarding the potential for validators and other users to exploit their power to capture what is colloquially referred to as *"maximal extractable value"* (MEV) through the censoring, swapping, and front-running of mempool transactions (Daian et al., 2020). In regimes where fee revenue accrues to the validator—quantity-control protocols, and any design without base-fee burning—I find that a block size target above roughly 37% of the demand-matching maximum leaves room for a monopolist validator to include her value-extracting transactions; under EIP-1559's base-fee burning, this fee-recycling channel is shut for myopic validators, and the residual discipline against MEV-motivated stuffing operates through the level of the burned base fee, which a lower target raises.

Blockchain designers need simple and robust economic insights to design their fee policies. With this need in mind, I conclude by suggesting open questions connected to this research.

1.1 Literature Review

The literature has generally explained blockchain fees arising because of competition for block space among heterogeneous, impatient users (Huberman et al., 2021). In the operations tradition, the pricing of congestible service systems goes back to Naor (1969) and Mendelson (1985), with Afeche and Mendelson (2004) characterizing priority pricing in queueing systems; the blockchain setting adds to this classical problem the interim monopoly of the validator and the protocol's commitment and measurability constraints. Easley et al. (2019) study the evolution of Bitcoin transaction fees, while Hinzen et al. (2022) highlight Bitcoin's limited adoption problem. Some studies highlight the strategic use of capacity on the Bitcoin blockchain (Lehar and Parlour, 2020; Malik et al., 2022). Catalini and Gans (2020) and Böhme et al. (2015) provide a simple primer on the economics of blockchains. Budish (2025) characterizes the economic limits of securing blockchains against attacks, which constrain the feasible supply of block space. This paper contributes to this literature by explaining the emergence of the two families of blockchain fee policies, those of Bitcoin and Ethereum.

Our modeling of demand uncertainty in blockchain usage and cryptocurrency investment is consistent with prior research. Aiello et al. (2023) documents the interaction of cryptocurrency price fluctuations with household investment behavior. Kogan et al. (2024) observe cryptocurrency usage and trades and identify momentum in cryptocurrency investments. In addition, fluctuations in uncertainty in Bitcoin block-production costs have been widely documented. Rehman and Kang (2021) show a negative correlation between the Bitcoin hash rate and energy commodity prices and Delgado-Mohatar et al. (2019) document increasing and varying electricity production cost of Bitcoin over time; these cost conditions bear on the marginal cost of block space through its per-transaction and orphan-risk components (Section 3.1). However, since the proof-of-stake upgrade, the marginal production cost of Ethereum blocks is essentially fixed, and participation as a validator is determined by stake. Jermann (2023) studies the steady-state dynamics of Ethereum stake. We use these facts to motivate our modeling approach based on demand and supply uncertainties for blockchains.

This paper contributes to the literature on price versus quantity controls, a domain pioneered by Weitzman (1974). The issue of choosing a supply function under uncertainty has been explored in Klemperer and Meyer (1989). My approach aligns closely with that of Reis (2006) and Flynn et al. (2026), who study the choice of a supply function from a firm's perspective and its macroeconomic implications. While my work draws inspiration from Weitzman (1974), it diverges in that it contemplates the planner's (blockchain designer's) problem with a variety of goals, including purely technical objectives like those seen in practice, such as block size targets. Weitzman (1974) studies a regulator choosing between taxing and capping a firm with an externality; in contrast, the sequential structure here—a designer commits to an instrument that a strategic agent then operates—shares the flavor of the strategic delegation literature, in which owners commit to managerial incentive schemes before product market competition unfolds (Fershtman and Judd, 1987; Sklivas, 1987). The blockchain setting differs from both traditions in that the agent is an interim monopolist selected by the consensus mechanism, the designer lacks coercive power and fiscal instruments, and the division of surplus is determined by bargaining between the protocol and its validators. The conclusions of this analysis are then applied to the design of transaction fee mechanisms (TFMs). Specifically, Ndiaye (2024) provides a summary of how the economic factors that affect the design of fee policies, as studied in this paper, correlate with the technical features of blockchains.

The literature taking a mechanism design perspective to examine fee policies is growing. Notably, Akbarpour and Li (2020) examine mechanisms immune to designer manipulations—referred to as "credible mechanisms"—and demonstrate that the well-known second-price auction does not meet this credibility criterion. In the blockchain context, Roughgarden (2020, 2021) applies this credibility condition to fee policies and establishes that EIP-1559, Ethereum's TFM, which essentially acts as a first-price auction with a dynamic reserve price, and its variations are incentive-compatible for users and adhere to a form of myopic credibility for validators (the full version is published as Roughgarden, 2024). Chung and Shi (2023) prove an impossibility result: no nontrivial TFM can simultaneously be incentive-compatible for users, incentive-compatible for validators, and robust to validator–user collusion; posted-price mechanisms emerge as essentially the only mechanisms satisfying both user and validator incentive compatibility. Ferreira et al. (2021) study dynamic posted-price mechanisms, and Bahrani et al. (2024) extend TFM design to settings with MEV. My contribution relative to this literature is distinct in three ways. First, whereas these papers characterize the incentive properties of a *given* mechanism, I solve the protocol's prior instrument-choice problem—price versus quantity controls—under demand and cost uncertainty, and derive when each regime is optimal as a function of bargaining power, elasticity, and the covariance of demand and costs. Second, I derive the optimal *dynamics* of the chosen instrument: the robustly optimal adjustment speed of an EIP-1559-style base fee equals the inverse price elasticity of demand, a parameter I then measure on Ethereum data. Roughgarden (2020) explicitly treats the base-fee update rule as exogenous and leaves its optimality outside his incentive analysis; that updating rule is precisely the object of my Section 4. Third, the incentive-compatibility results of this literature serve as inputs to my welfare analysis: they pin down when the truthful-bidding benchmark applies exactly, and Section 3.2 analyzes how strategic bidding under quantity controls affects my comparison.

This paper also advances the literature examining the block space market from a macroeconomic perspective. The concepts formalized in Section 3 of this paper build upon and extend Buterin's (2018) reading of Weitzman (1974), with important differences characterizing the blockchain context because the protocol designer is limited in her capacity to enforce quantity or price controls. In other related work, Lavi et al. (2022) and Nisan (2023) investigate the monopolistic market for unlimited block space, in contrast to this paper, which takes the block size limit as an exogenous technological constraint.

This paper also relates to a growing empirical literature in economics and finance on blockchain fee mechanisms, market structure, and inclusion. Makarov and Schoar (2021) document substantial concentration in Bitcoin mining and ownership. Cong et al. (2023) provide early evidence from the Ethereum ecosystem on whether Web3 and decentralized finance democratize participation, documenting concentration among miners and holders, congestion-driven spikes in gas fees that disproportionately exclude smaller users, and the redistributive consequences of EIP-1559's base-fee burning and of airdrops. Liu et al. (2022)

empirically evaluate EIP-1559 and find that it improves the predictability of user bids and reduces waiting times, while fee levels remain primarily demand-driven. My theory complements this empirical evidence rather than restating it: taking as given the documented facts—concentrated block production, volatile congestion-driven fees, and the burning of base-fee revenue—I provide a framework that explains *why* a protocol facing these conditions optimally adopts a price-control regime, *how fast* its base fee should adjust, and *how large* the block-size buffer must be to deter validator self-dealing. Conversely, the welfare costs of misallocated block space that motivate my designer's objective are precisely the congestion-exclusion effects that this empirical work documents.

Last, this paper contributes to studies of the dynamics of the Ethereum fee policies recently adopted in EIP-1559. Leonardos et al. (2021) studies the behavior of the dynamic system resulting from the fee policies, and Leonardos et al. (2023) uncovers numerous empirical properties for which this paper provides a theoretical explanation.

Outline: The paper is organized as follows: Section 2 gives an overview of the functioning of the Bitcoin and Ethereum protocols, makes the case that validators have had more bargaining power in the history of Ethereum than in that of Bitcoin, and explains how Ethereum fees are determined. Section 3 introduces the model, analyzes incentive compatibility and strategic bidding under both regimes, provides the main results of my analysis, discusses the role of the key modeling assumptions, and presents an extension to cryptocurrency price fluctuations. Section 4 studies the dynamics of fee policies and their implications for Ethereum. Section 5 distills the design guidance and managerial implications of the analysis. Last, Section 6 concludes the paper.

2 Transaction Fees and Block Space Utilization in Bitcoin and Ethereum Protocols

2.1 The Evolution of the Bitcoin and Ethereum Protocols and Fee Policies

A blockchain is a decentralized digital ledger that records transactions across a network of computers, called validators, in an immutable chained list of blocks. Bitcoin, introduced by Nakamoto (2008), was conceived as a peer-to-peer digital currency. In contrast, Ethereum, proposed by Buterin (2013), extends the basic blockchain concept to include more versatile functionalities such as self-executing agreements, known as "smart contracts".

In Bitcoin and Ethereum, each block has a fixed capacity because of technological constraints such as bandwidth and storage and the negative externalities associated with large blocks' propagation times. These limitations ensure that running a node remains accessible to a broad range of participants, thereby fostering decentralization. This limited capacity necessitates transaction fees, which serve as a market mechanism to allocate this scarce resource. Fees incentivize validators to prioritize and include transactions in a block.

Over the years, both Bitcoin and Ethereum have experienced changes in their block capacity and fee structures to adapt to changing network demands. On the price side, Bitcoin shifted from offering no-fee transactions to imposing transaction fees, as studied by Easley et al. (2019). On the block capacity side, since its inception, Bitcoin has given control of this capacity to the protocol developers. The original 1MB block size limit in the Bitcoin blockchain ignited intense debates within the community over scalability versus decentralization. On one side, proponents of a larger block size argued that increasing capacity would enable more transactions per block, alleviating congestion and lowering fees. On the other side, critics warned that larger blocks would raise the computational and storage requirements for running a validator, compromising the network's decentralization. This ideological divide peaked in 2017, leading to a "hard fork" that birthed Bitcoin Cash, a separate chain with an 8MB block size. Concurrently, Bitcoin adopted Segregated Witness (SegWit), effectively changing the block size limit to a more complex "block weight" limit of approximately 4 million units.

Ethereum, for its part, imposed transaction fees from the beginning. In Ethereum, "gas" serves as the unit of resources, and the "gas limit" dictates the maximum network capacity in a single block. Initially, Ethereum attempted a different paradigm and gave control over capacity to validators. Each validator was allowed to change capacity up or down by 0.1% for each new block. The underlying rationale was to transform the philosophical debate on block size into an economic decision, assuming that validators, heavily invested in the network, would act individually in the long-term interest of the network. However, in practice, the history of Ethereum block size changes in Table A1 in Appendix A illustrates that validators have often abused their bargaining power and colluded to manipulate network capacity. This behavior is exemplified by an April 21, 2021, announcement from Sparkpool, a major validator based in China with 23.5% of the network's computational power at that time:

```
@sparkpool_eth:  "We are raising gas limit to 15 million"
@btcdentist:  "did you get permission from the core devs?"
@sparkpool_eth:  "What we need is advice from dev, not permission."
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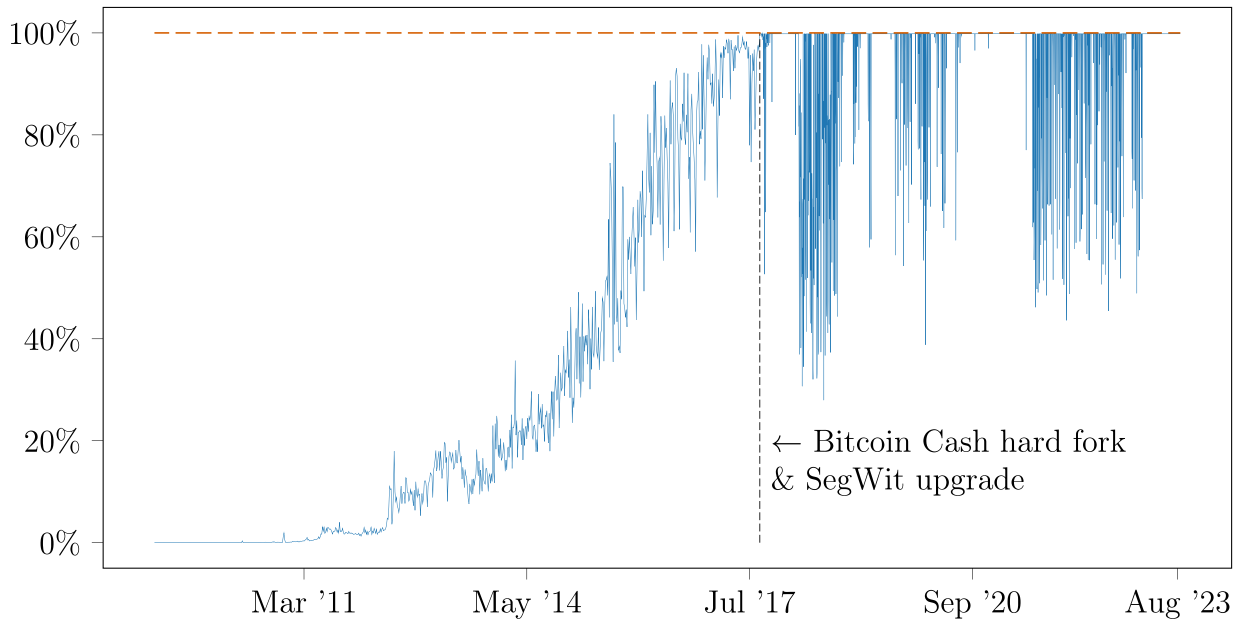
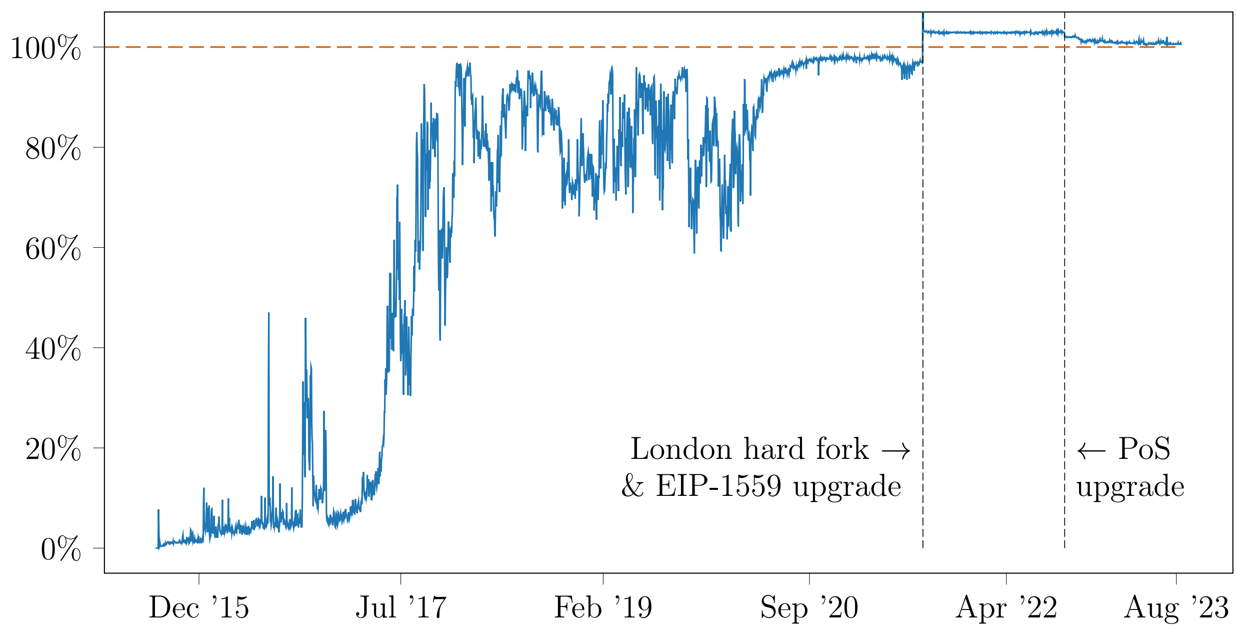
To curb such practices, Ethereum introduced EIP-1559, a fee mechanism designed to limit validators' discretionary power over network capacity and enhance user fee predictability.



(a) Bitcoin capacity utilization rate per day



(b) Ethereum capacity utilization rate per day

Figure 1: Time series of Bitcoin and Ethereum capacity utilization rates

Figure 1 illustrates the time series of the capacity utilization rates of the Bitcoin and Ethereum blockchains. Considering blockchain upgrades over time and their variations, I define the block size target as the size below which the reserve price for pending transactions does not increase. The capacity utilization rate is, consequently, the daily average of the fraction of the realized block size to the block size target.

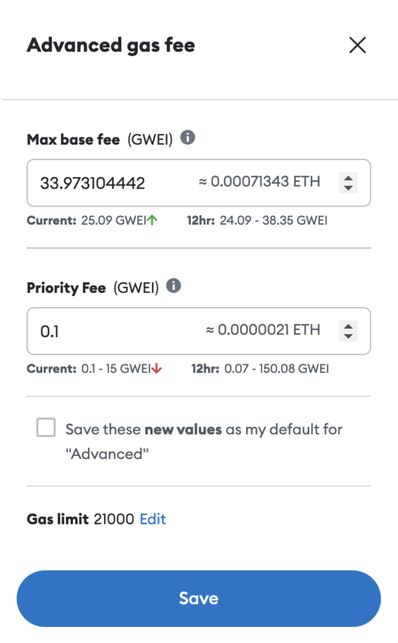
For Bitcoin, the block size target was 1MB pre-SegWit and is 4 million weights post-upgrade. The top panel indicates an initial surge in Bitcoin's utilization rate during its nascent phase. However, as its use has become more widespread, the network has not continuously operated at full capacity. This fact is corroborated by Lehar and Parlour (2020), who attribute the prevalence of less-than-full blocks to strategic capacity management by monopolistic validators.

Before the London hard fork, Ethereum's block size target varied based on validator votes, ranging from 3.1 million to 15 million gas, as detailed in Appendix A. Post–London fork and EIP-1559 implementation, the target is deterministic and equal to half the gas limit—15 million gas against a 30 million limit at implementation, doubled to 30 million against 60 million by validator votes in 2025 (see below)—with a dynamic reserve price adjustment for pending transactions to align closely with the target. The bottom panel reveals that, just as for Bitcoin, Ethereum's early-phase utilization rate rose steadily. However, following the introduction of EIP-1559 via the London hard fork, the blockchain has operated at full capacity. My analysis attributes this to EIP-1559's design, arguing that transaction fees now effectively shape and regulate the supply curve for monopolistic validators.

Ethereum's fee mechanism has continued to evolve since EIP-1559, in ways worth keeping in view for the analysis that follows. The Dencun upgrade (March 2024) implemented EIP-4844, which introduced "blob"-carrying transactions for rollup data with a *separate* fee market: the blob base fee is an exponential function of the cumulative excess of blob usage over a deterministic target, with the update fraction chosen so that the price changes by at most a factor of $e^{0.118} \approx 1.125$ per block, matching EIP-1559's 12.5% cap (Buterin et al., 2022). The blob market is thus a second instance of the adaptive-to-target family of fee policies whose dynamics this paper studies in Section 4—and, notably, it adopts the exponential adjustment-function shape that my analysis derives as robustly optimal (Proposition 4), while inheriting an adjustment rate calibrated to EIP-1559's cap rather than to the elasticity of demand. Subsequent upgrades scaled the parameters of both fee markets: Pectra (May 2025) raised the blob target and maximum (EIP-7691) and the cost of calldata (EIP-7623); validators raised the gas limit from 30 million to 60 million over the course of 2025; and Fusaka (December 2025) introduced data-availability sampling (EIP-7594), a reserve price tying the blob base fee to the execution base fee (EIP-7918), and a lightweight fork mechanism for adjusting blob parameters (EIP-7892) (Nijkerk, 2025; Ethereum Foundation, 2025). Throughout these changes, the structure analyzed in this paper—a posted, rule-adjusted price with quantity flexing around a deterministic target—has remained the organizing principle of Ethereum's fee design.

Such economic interactions between fee mechanisms and capacity have been overlooked in earlier studies. For instance, Lehar and Parlour (2020) focus on strategic capacity utilization in Bitcoin, while Huberman et al. (2021) theoretically ascribe full capacity utilization to the free entry of validators even though Bitcoin does not continuously operate at full capacity despite such entry. In my analysis in Appendix B.1, I examine variables related to block space demand: exogenous demand shifters—the number of active addresses and token prices—together with two endogenous congestion outcomes, transaction fees and (for Bitcoin) residual demand in the mempool, which proxy for demand conditions rather than shift them. The data reveal strong correlations between these demand-side variables and the utilization rate for Bitcoin and Ethereum pre-EIP-1559. Post-EIP-1559, these factors became decoupled from block utilization. Two caveats apply to this before/after comparison: the stabilization of utilization at the target is the mechanical implication of the base-fee rule operating as designed rather than an estimated treatment effect, and the post–EIP-1559 period coincides with the adoption of Layer-2 rollups and pronounced swings in cryptocurrency markets, which independently shift mainnet demand. I therefore read the decoupling as evidence on the mechanism's operation, not as a causal estimate of EIP-1559's effect on demand.

In my study on block space supply in Appendix B.2, I evaluate factors such as computing power and mining pool concentration for both Bitcoin and Ethereum, adding the total Ethereum staked and validator pool concentration for Ethereum post–proof of stake. The results indicate no discernible association between computing power (or the amount of Ethereum staked) and block utilization rates for Bitcoin or Ethereum; because both series exhibit strong secular trends over the sample, I read these plots as descriptive rather than causal. However, higher concentration rates among mining pools in Bitcoin, measured through the Herfindahl–Hirschman index (HHI), correlate with sub-100% block utilization, suggesting that miners may exercise market power to leave blocks less full. This pattern does not hold for Ethereum, where both before and after the transition to proof of stake, mining pool concentration shows no discernible association with block utilization rates. The stabilization of Ethereum's block utilization post–London fork further corroborates this observation.



(a) User price choice from Metamask wallet

(b) Base fee and priority fee estimator from Blocknative

Figure 2: Ethereum fee settings and estimator

2.2 Transaction Fees on Ethereum

Ethereum fees initially operated on a bidding system constrained by a fixed block size. The fixed per-block gas limit and fluctuations in demand resulted in user delays, as the system lacked a "slack" mechanism to adjust block size to meet varying demand. In addition, since validators collected the fees, this system led to a first-price auction when demand was high. As first-price auctions are not incentive-compatible when user valuations are unobserved, this required complex fee estimation efforts from users.

With EIP-1559, Ethereum's fee structure underwent significant changes to address the issues with the first-price auction with a fixed block size limit. The system was implemented with a target block size set at $q^{target} = 15M$ gas, a legacy from the pre–London hard fork settings, and a maximum block size of $q^{\max} = 2q^{target}$.[6] Each transaction comes with a "base fee" on the user end, algorithmically adjusted based on network demand. The minimum gas

[6] These were the values at EIP-1559's implementation and throughout the sample period studied in Section 4. Validators doubled the gas limit to 60 million over the course of 2025 (Section 2.1); because the target adjusts one-for-one with the limit, the ratio $q^{\max}/q^{target} = 2$, the object relevant for the analysis, is unchanged.

price, p_t, is adjusted based on the formula

$$p_t = p_{t-1} \cdot (1 + d\frac{q_{t-1} - q^{target}}{q^{target}}) \tag{1}$$

where d is an adjustment parameter set to $\frac{1}{8}$, so that the base fee rises by 12.5% per block when blocks are full and falls by 12.5% per block when they are empty. In addition to the base fee, users can include a "tip" to incentivize faster processing by validators. The dual-fee structure allows more predictability in transaction costs, as the base fee aligns closely with network congestion. Figure 2 shows fee settings and an estimator of the base fee for Ethereum users.

From a validator's standpoint, selection is stochastic, contingent on either the proof-of-work or proof-of-stake mechanism. Though the selected validator enjoys a monopolistic position during her turn to validate a block, the protocol's fee structure regulates her behavior. Each transaction j carries a computational cost q_j, measured in gas units. Ethereum transaction senders pay an amount computed as $q_j \cdot \min\{p_t + \delta_j, c\}$, where δ_j is the tip and c the fee cap, with $c \geq p_t$. Incentive compatibility requires that, under normal demand conditions, the base fee adjusts upward to match the willingness to pay of the marginal user, and the tip is small in proportion. The aggregate base fee revenue over N transactions, $\sum_{j=1}^{N} q_j p_t$, is "burned", primarily to address the validator's off-chain incentives, as argued by Roughgarden (2021). Meanwhile, tips and a block reward go directly to the validator. By diverting a portion of the revenue away from the validator, the protocol has instruments to ensure that supply is not artificially restricted and that transactions of higher value are included.

EIP-1559 raises several questions regarding its rationale, potential for improvement, and the design of other fee policies that share its simplicity. Specifically, in the face of demand fluctuations, why might a protocol designer opt for a blockchain with an imposed quantity limit as in Bitcoin and Ethereum pre–London hard fork and dynamically adjust when demand fluctuates vs. one where price controls such as the EIP-1559 base fee are imposed and dynamically adjust when demand fluctuates? What determines the shape and parameters of such transaction fees? These questions will be the focus of subsequent sections in this paper.

3 Model

3.1 Environment

A *blockchain*, such as Bitcoin or Ethereum, is modeled as a distributed computer network where users submit *transactions* to be included in a chain of *blocks* by *validators*. The blockchain records the network's state, such as account balances. A transaction t represents arbitrary data sent over the network to alter its state—for instance, to transfer a balance. Users submit transactions to a publicly observed pool of outstanding transactions (*mempool*), with a bid b_t, signifying their willingness to pay for transaction processing. The monopolistic validator, using quantities x_i of a finite number of resources $i \in [\![1, N]\!]$ (e.g., computation, bandwidth) selects a subset of transactions from the mempool to form a block. A block of size q is an ordered sequence of transactions and a reference to the previous block. Validators add a block to the blockchain by a consensus mechanism (such as proof of work or proof of stake), a process irrelevant to this analysis. Technological constraints impose a maximum block size $q^{\max}$, thus limiting the supply of block space.[7]

Validators: We model a setting with multiple validators; however, when a validator is selected to produce a block, she acts as a monopolist. This fact reflects the temporary monopoly power inherent in blockchain consensus mechanisms with a single block proposer. The validator uses a bundle of computational resources $x \in \mathbb{R}^N_+$ at exogenous per-unit resource rates $g_x \in \mathbb{R}^N_{++}$ to produce a block of size $q \leq q^{\max}$, as given by

$$q = \sum_{i=1}^{N} g_{x_i} x_i \tag{2}$$

Each transaction consumes specific amounts of various resources (e.g., computation, storage, bandwidth). The total block size is the weighted sum of these resource usages across all included transactions.

The validator incurs a cost $C(q) = c(x_1, \ldots, x_N)$, based on the resources used to produce a block. These costs encompass validation operation costs and other costs associated with accessing and modifying the blockchain's state.[8] Denote by $C(q; \eta)$ the validator costs of production, where η is a random variable with finite mean that represents the validator's marginal cost uncertainty.

Two features of this cost structure matter for the analysis. First, the cost of production

[7] See the related discussion in Buterin (2018) for the case of the Ethereum blockchain.

[8] From the blockchain designer's viewpoint, costs might also include block propagation delays due to large blocks and other societal costs.

depends on the size of the block. Including an additional transaction consumes computation (gas metering and signature verification), bandwidth, and storage for state access, and—under proof of work—raises propagation time and thus the risk that the block is orphaned, an expected loss proportional to the value at stake in the block. These components scale with block contents. The consensus-layer costs of participating—hashing in proof of work, staking in proof of stake—are instead independent of which transactions a block contains: they are fixed costs from the standpoint of the block-composition decision, so they do not affect the marginal conditions that determine prices and quantities; they enter the analysis through the validator's participation, which the bargaining problem of Section 3.4 captures through the requirement that validators retain sufficient excess profits.

Second, the marginal cost of block space varies and is uncertain, and the shock η captures this variation at the policy-relevant horizon. The model is one of a stationary environment in which the protocol commits to a fee policy ex-ante, before the realization of the shocks (λ, η); η should accordingly be read as the realization of cost conditions over weeks to months—the frequency at which energy prices and network conditions vary and at which protocols revisit their fee rules—rather than as block-by-block noise. For proof-of-work systems, the marginal cost of producing electricity varies systematically with the season, and at higher frequency within the day (Borenstein and Holland, 2005); diurnal fluctuations average out over the policy horizon, so it is the seasonal and cyclical components of power prices that constitute the persistent cost conditions η represents. The representative validator's problem is the per-block problem of the interim monopolist, evaluated at those prevailing cost conditions; because validators are selected repeatedly from the same pool, $C(q; \eta)$ can equivalently be read as the flow cost of producing blocks of size q in a stationary equilibrium. As Section 3.4 shows, what matters for the choice of fee policy is not only this uncertainty but also its covariance with demand.

Users: Users, denoted by $j \in [0, 1]$, are atomistic. We model user arrivals between two consecutive blocks, B_t, B_{t+1} as a Poisson process X with rate ζ. To facilitate comparative analysis under different block size limits, we introduce the notation $\zeta \equiv \lambda q^{\max}$, where λ reflects the arrival rate per unit of maximum block size. In our subsequent analysis, we consider λ a random variable capturing demand uncertainty. We work throughout with the continuum (large-market) limit of the arrival process, identifying realized demand at price p with its intensity $\lambda q^{\max} \bar{F}(p)$ rather than with the finite Poisson count; the remark in Appendix C.4 discusses the finite-bidder case. For simplicity, we assume that users leave the pool if their transaction is not included in the next block, only to return according to the arrival process.[9]

Each user j has a valuation v_j drawn from a common distribution f with a cumulative distribution function F, which is continuous and increasing.

Establishing a Demand Curve: The baseline analysis assumes that users bid their true valuations, i.e., $b_t = v_j$. Section 3.2 examines when this assumption holds exactly—it does under the price-control mechanisms studied here, as demonstrated by Roughgarden (2020, 2021)—and shows that the paper's welfare comparison is robust to strategic bidding under quantity controls. Under truthful bidding, for a given minimum bid for transaction inclusion, denoted by p, the number of users willing to pay the bid is $\lambda q^{\max}\bar{F}(p)$, where $\bar{F}(p) = 1 - F(p)$. The following lemma shows that this model is the microfoundation of an intuitive demand curve for block space, thereby linking demand parameters to model primitives.

Lemma 1. *The aggregate demand for block space can be represented as* $p = \left(\bar{F}\right)^{-1}\left(\frac{q}{\Psi}\right)$, *where the price elasticity of aggregate demand, evaluated at the valuation* p *of the marginal included user, equals the tail ratio* $\frac{pf(p)}{1-F(p)}$.

Specifically, when F *is a Pareto distribution with scale* p_m *and shape* α, *the aggregate demand for block space is given by*

$$\frac{p}{p_m} = \left(\frac{q}{\Psi}\right)^{-\frac{1}{\varepsilon}} \tag{3}$$

Here, $p \in \mathbb{R}_+$ *is the market price,* $\Psi \equiv \lambda q^{\max}$ *is a demand shifter, and* $\varepsilon = \alpha$ *is the price elasticity of demand for block space.*

Proof. Refer to Appendix C.1 for the proof. □

In Section 3.4, assume that users' valuations for transaction inclusion follow a Pareto distribution. This choice is motivated by empirical observations in blockchain networks, where transaction fees often exhibit heavy-tailed distributions due to a small number of users willing to pay significantly higher fees for urgent processing. For instance, data from the Ethereum network shows that transaction fees follow a heavy-tailed distribution (Guo et al., 2019, See), justifying the Pareto assumption.

While the Pareto distribution is a simplification, it allows for analytical tractability. The assumption of a Pareto distribution is not restrictive. For a general user valuation distribution, one can calculate the price elasticity of demand at all points from the tail ratio of the distribution.

[9] Section 3.2 discusses this assumption, user participation, and the robustness of fee dynamics to residual mempool demand.

The demand curve from the Pareto distribution will be helpful when I consider uncertainty in the user arrival rate λ, leading to uncertainty in the demand shifter Ψ. By modeling λ as a random variable, we incorporate demand uncertainty into our analysis and can study the effect of stochastic demand on the optimal fee policies. For realization $\lambda > 1$, we encounter a high-demand scenario where not all transactions can be included in the next block, whereas $\lambda < 1$ reflects a low-demand scenario where the block is not filled. Given this context and considering the uncertainties in both the cost $C(q; \eta)$ and demand Ψ, we study the conditions under which a blockchain protocol designer would find introducing price or quantity controls beneficial.

3.2 Incentive Compatibility and Strategic Bidding

The demand curve in Lemma 1 is built on truthful bidding. This subsection examines the game-theoretic foundations of that assumption under each fee regime and establishes that the paper's welfare comparison survives when users bid strategically. The two regimes are not symmetric in their incentive properties, and this asymmetry is worth making explicit.

Price Controls: Under the price-setting regime, the protocol posts a base fee p, the base-fee revenue is burned, and the validator fills the block with the transactions that pay it. This is a posted-price mechanism: a user's bid determines only whether her transaction is included, not the price she pays at the margin. Roughgarden (2020, 2021) prove that, except in periods of rapidly growing demand in which the base fee is below the market-clearing level, EIP-1559-type mechanisms are dominant-strategy incentive-compatible for users—the optimal bid is one's true valuation (a fee cap equal to the valuation with a minimal tip)—and incentive-compatible for myopic validators, because burning the base fee removes the validator's incentive to stuff blocks with fake transactions to inflate future fees. Truthful bidding under price controls is therefore not an assumption but an equilibrium property.

Quantity Controls: Under the quantity-setting regime, the protocol fixes the block size and fees are determined by competitive bidding—in practice, a pay-as-bid (first-price) auction, as on Bitcoin. First-price auctions are not dominant-strategy incentive-compatible: inframarginal users optimally shade their bids toward the expected market-clearing price. This is not an artifact of Bitcoin's particular design that a cleverer quantity-control mechanism could avoid. Chung and Shi (2023) prove that no nontrivial TFM can simultaneously satisfy user incentive compatibility, validator (miner) incentive compatibility, and robustness to validator–user side contracts; and mechanisms in which the validator keeps the fee revenue—as a quantity-control regime requires if fees are to compensate validators—cannot be user-DSIC except through posted prices, which is precisely the price-control regime (Ferreira et al., 2021). In other words, within the quantity-control regime, some departure from truthful bidding is unavoidable. Second-price formats do not help: they are not credible for a monopolist auctioneer (Akbarpour and Li, 2020), which is why they are not used as TFMs.[10] The relevant question is therefore not whether users bid truthfully under quantity controls—they do not—but whether strategic bidding overturns the welfare comparison between the regimes. The following result shows that it does not.

Proposition 1. *Consider the quantity-setting regime with block size q, in which included transactions are selected by highest bids and users pay their bids. Suppose users are atomistic and observe the state of the public mempool (equivalently, the realized demand state Ψ) before bidding, and consider a realized state in which block space is scarce, $q < \Psi$. Then there exists a Nash equilibrium in which every user with valuation $v_j \geq p^*$ bids the market-clearing price $p^* = p_m(q/\Psi)^{-1/\varepsilon}$ and every user with valuation $v_j < p^*$ abstains. In this equilibrium, the set of included transactions, the realized block size, and the validator's fee revenue p^*q coincide with their values under the market-clearing benchmark of Lemma 1, in which included users pay the clearing price—the outcome that truthful bidding delivers under price controls. Consequently, on the scarcity states—those in which the fee policy binds—the quantity-control outcome coincides state by state with this benchmark, leaving the value of quantity controls and the welfare comparison between the two regimes (Proposition 2 below) unchanged.*

Proof. Refer to Appendix C.4. □

The logic is that of price-taking in a large market: no individual user is pivotal for the clearing price, so bidding above p^* only increases one's payment, while bidding below p^* forfeits inclusion. Strategic behavior redistributes surplus from the validator's auction rents to users relative to a hypothetical pay-as-bid auction with truthful bids, but it leaves the allocation and the revenue at the competitive benchmark that the model uses. Two caveats delimit this robustness result. First, it relies on users observing the demand state through the public mempool; when demand is observed with noise, bids shade relative to the realized clearing price and outcomes converge to the competitive benchmark only as the market grows large, in the sense of Swinkels (2001). The comparative statics of the regime comparison (Proposition 2 below) are then a limiting approximation under quantity controls, while they remain exact under price controls—if anything, this asymmetry favors price controls

[10] A hybrid design could post a price *and* fix quantity, rationing at the posted price; this forgoes the allocative role of fees—rationing no longer selects the highest-value transactions—and corresponds to neither Bitcoin's nor Ethereum's design, so we restrict attention to the two pure regimes.

beyond what Proposition 2 captures, since price controls do not rely on users forecasting the clearing price. Second, the validator's own incentives under quantity controls—to leave blocks partially empty or to fill them with her own value-extracting transactions—are a separate concern from user bidding; Section 4 derives bounds on the block-size target under which complying with the target is incentive-compatible for a myopic validator.

User Participation and Resubmission: The model assumes that users whose transactions are not included exit the mempool and return only through the arrival process. Interim participation is never violated: in both regimes, an included user pays at most her valuation, so no user regrets transacting. What the exit assumption rules out is intertemporal substitution—users queueing in the mempool across several blocks. Leonardos et al. (2021) confirm that this exit assumption does not significantly affect base-fee dynamics, and Nisan (2023), who accounts for residual demand in the mempool, obtains fee dynamics similar to those in Leonardos et al. (2023). Queueing matters for one quantitative object: it makes demand for *next-block* inclusion more elastic, since waiting is a substitute for bidding up. This is precisely why the elasticity in Section 4 is defined as the price elasticity of inclusion in the next block, and why the measured value is high; the implied optimal adjustment rate is correspondingly low. A formal queueing treatment would go further: with a persistent mempool, demand for next-block inclusion is the flow of new arrivals plus the backlog of waiting users, whose option to keep waiting disciplines bidding—the structure of Huberman et al. (2021), who model blockchain fees as a queueing system. Two implications are worth recording. First, the elasticity relevant for a per-block adjustment rule remains the next-block inclusion elasticity, which queueing raises; since the optimal adjustment rate is its inverse, accounting for queueing pushes the optimal rate further down, reinforcing the conclusion of Section 4 that Ethereum's current rate is too fast. Second, a backlog makes effective demand serially dependent, which preserves the law of motion of Proposition 5(i)—whose innovation is realized demand growth of any form—but would modify the stationary-variance formulas; deriving the optimal adjustment rate in a full queueing model is a natural next step.

3.3 Tension Between Protocol Objectives and The Validator's Incentives

Given that each validator has monopoly power during their assigned turn, we model the interaction between a single validator and the protocol designer. This allows us to analyze the strategic choices and conflicts in setting fee policies.

3.3.1 Protocol Designer's Preference for Maximum Capacity Utilization

The designer selects the control variable (price or quantity) before uncertainties are realized, aiming to maximize expected social welfare while considering the strategic responses of validators and users.

If the protocol designer has a preference over the block size, then she will prefer quantity controls over price controls as the former insulates this variable from demand fluctuations. To illuminate the potential conflict between the protocol designer's objectives and the miners' incentives, consider the following examples of objective $S(q)$ for the protocol designer.

Example 1. *(Social Welfare) Let u be an increasing function representing the utility of a representative user on the blockchain. The social welfare function $S(q)$ can be defined as*

$$S(q) = \begin{cases} u(q) & \textit{if } q \leq q^{\max} \\ -\infty & \textit{if } q > q^{\max} \end{cases} \tag{4}$$

This example reflects a protocol designer's concern for maximizing user utility, an objective that can diverge from the profit-maximizing motive of the validator.

Claim 1. *Suppose the protocol designer aims to choose either a base fee (price control) or a block size limit (quantity control) to maximize social welfare as defined in Equation (4). Then (i) the quantity control $q^{\max}$ attains the maximal feasible block size, $\min\{\Psi, q^{\max}\}$, in every demand state, while any price control $p > p_m$ entails a deadweight loss in low-demand states; and (ii) in any demand state $\Psi = \lambda q^{\max}$ with $\lambda < 1$, no price above $p_m \lambda^{1/\varepsilon}$ fills the block—along the isoelastic extension of the demand curve, that price exactly fills it—and this threshold vanishes as demand intensity vanishes, so no fixed positive price keeps blocks full across all low-demand states.*

Proof. Refer to Appendix C.2. □

Setting a sufficiently low price eliminates the deadweight loss from less-than-full blocks in low-demand conditions, and the required price vanishes as demand intensity does. This claim highlights a point of conflict with the validator, who may have different objectives, such as profit maximization.

Example 2. *(Technological Block Size Targets) The protocol designer may have a target block size q^{target} to optimize network performance, security, or decentralization. For example, setting a target block size can help manage the trade-off between transaction throughput and the time it takes for blocks to propagate through the network. She might also allow some degree of deviation from the target block size, which we can represent as $S(q) = \ell\{q - q^{target}\}$, where*

ℓ stands for some loss function. For instance, a square loss function could be used to penalize deviations from the target:

$$S(q) = -\left(q - q^{target}\right)^2 \tag{5}$$

Claim 2. *Under a technological block size target, the protocol designer prefers quantity controls over price controls. In particular, for the square loss function, the expected block size under optimal price controls $\mathbb{E}[q]$ is less than the target q^{target}, and the loss is given by*

$$\mathbb{E}[\left(q(\Psi) - q^{target}\right)^2] = \frac{Var(\Psi)}{\mathbb{E}[\Psi^2]}(q^{target})^2 \tag{6}$$

Proof. Refer to Appendix C.3. □

This loss quantifies the variance in the block size, relative to the target, that arises because of the fluctuations in demand Ψ when the price is optimally chosen ex-ante. This highlights that the loss for the protocol designer is more significant when the demand shifter has a high variance.

3.3.2 The Validator's Monopoly Power

While validators compete ex-ante to be selected for block production, the selected validator effectively becomes a monopolist for that block. This model captures the essential features of the validator market: competition determines which validator gets to produce a block, but the selected validator holds monopoly power at the interim stage. A monopolist validator's profit Π consists of the block reward R, the transaction fee revenue pq, and the cost $C(q;\eta)$ associated with producing a block of size q.[11]

$$\Pi = \mathbb{E}\left[R + pq - C(q;\eta)\right]. \tag{7}$$

A note on fee incidence is in order. We write the validator's variable revenue as the full market value pq of included block space. Under fee retention—Bitcoin, and Ethereum before

[11]For simplicity, I abstract from the fact that not all fees collected go to the validator, and the monopolist validator can capture extra revenue from MEV. In Ethereum's EIP-1559 mechanism, the base fee is burned rather than paid to validators. Hence, validators only receive the priority tip component of users' payments. This burning mechanism is critical: without it, validators could insert empty transactions to inflate future block fees at no net cost, undermining the price-adjustment process. In addition to transaction fees, validators can capture MEV by reordering or selectively including transactions to exploit on-chain arbitrage opportunities (e.g., DEX front-running). On Ethereum, a sophisticated ecosystem of searchers and builders helps validators systematically capture such profits.

EIP-1559—this is the literal cash flow. Under EIP-1559, part of users' payments is burned, and the validator's cash receipts consist of tips and MEV; the footnote above abstracts from this, and the ϕ-scaling discussed in Section 3.4 shows how receipts can be rescaled, with deterministic rescalings leaving the instrument comparison unchanged because they shift fee receipts identically in the two regimes. The bargaining analysis should accordingly be read as bargaining over the gross surplus that block space generates—with burning determining how much of that surplus is destroyed rather than transferred—while the incentive analysis of Section 3.2, which turns on the cash incidence of the base fee, treats burning explicitly. A fully articulated incidence model tracking user payments, burned surplus, and validator transfers state by state is a natural extension of the framework. Application developers and users design transactions to be as efficient as possible to minimize costs, i.e., the cost to the validator is[12]

$$C(q; g_x, \eta) = \min_{x \in \mathbb{R}_+^N} c(x_1, \ldots, x_N; \eta) \tag{8}$$

subject to (2). The bundle x can be interpreted as the various resources that constitute a user transaction, such as bandwidth and computational operations. Let us consider that c is homogeneous of degree 1 in x.[13] We then have

$$C(q; g_x, \eta) = \Gamma(\eta, g_x) q \tag{10}$$

The expression for the marginal cost Γ is derived in Appendix C.6.

A monopolist validator would then maximize profits in equation (7). In contrast to the protocol designer, who systematically prefers to set quantities ex-ante, the monopolist validator might choose either the price-setting or the quantity-setting regime, depending on the degree of demand uncertainty.

3.4 Bargaining Problem

I model the resolution of the potential conflict between the protocol designer and the validator by assuming that the price- or quantity-setting regime is determined by Nash bargaining over

[12]In blockchain networks, transaction fees are often proportional to the computational and storage resources required. Therefore, developers and users are incentivized to optimize their transactions to reduce costs.

[13]A typical example is the Cobb–Douglas cost function

$$c(x_1, \ldots, x_N; \eta) = \eta \prod_{i=1}^{N} x_i^{\varepsilon_i} \quad \text{such that} \quad \sum_{i=1}^{N} \varepsilon_i = 1 \tag{9}$$

the protocol profits.[14] Before uncertainty is realized, the protocol designer and the validator commit ex-ante to a fixed base fee (price-setting) or fixed block space (quantity-setting). In practice, such a process is implemented by a set of rules in the fee policies of the blockchain to which validators and blockchain designers agree. Even though the validator has ultimate control over which transactions, of what size, and at what price can be included in a block, sophisticated mechanisms can be encoded to enforce prices and quantities. Protocol revenue can be diverted in part to the protocol treasury, burned, or rebated to users to implement the fee policies; this design feature is beyond the scope of this paper.[15] The planner's objective balances social welfare and technological considerations while ensuring that the validator has enough excess profits to be willing to provide her services.

The following objective captures the planner's problem:

$$\mathcal{V} = \mathbb{E}\left[S(q)\right]^{1-\beta}\,\mathbb{E}\left[pq - C(q;\eta)\right]^{\beta} \tag{11}$$

In this objective, the parameter $\beta \in [0;1]$ captures the bargaining power of the validator. $\beta = 1$ means that the outcome maximizes the excess profits of the validator,[16] while $\beta = 0$ means that the outcome optimizes for the protocol designer's objective captured by the function $S(q)$. The Nash bargaining problem is defined over expected utilities, taking into account uncertainty in demand λ and marginal costs η. We assume that both parties have common knowledge of the distributions of these random variables.

Three remarks on this objective are in order. First, on why validator profits enter the objective at all, rather than appearing only through incentive or participation constraints as in standard mechanism design: the protocol designer is not a principal with enforce-

[14]Even without a formal bargaining game, one could analyze the fee mechanism preferences of the interim monopolist validator. The main takeaways on price vs. quantity-setting can stand alone as a simpler exercise.

[15]See Roughgarden (2021).

[16]For analytical simplicity, we focus on transaction fees as the primary variable component of the validator's profit. While the block reward is important, it is a fixed amount per block in both Bitcoin and Ethereum—it does not vary with block contents—and therefore does not directly influence the choice between price and quantity controls in our model. A *state-contingent* block reward—one that rises when demand is low—would be an additional instrument for the designer: it could subsidize validator participation in low-demand states and relax the tension between the designer's and the validator's objectives. No major protocol implements state-contingent rewards, in part because any on-chain measure of the demand state (e.g., mempool size) is unrecorded and manipulable by the validator herself, which is why we restrict attention to the instruments observed in practice. Another interpretation of our analysis is to look at the long run, where block rewards are often programmed to decay exponentially to zero in order to avoid overinflation; Bitcoin's halvings make fees the dominant margin precisely over the horizon for which fee policy matters most. An analysis involving the full profit of the validator $\mathbb{E}\left[R + \phi pq - C(q;\eta)\right]$, where ϕ is the stochastic ratio of the validator's fee receipts to the posted price p—values below one capture the burning of part of users' payments, and values above one a priority-fee surcharge collected on top of p—as in Jermann (2023), would yield similar insights. A deterministic ϕ rescales fee receipts identically in both regimes and leaves the comparison $\Delta^{\log}$ below unchanged.

ment power. She cannot tax validators, cannot verify their costs, and cannot compel them to process transactions; the protocol's continued operation rests on validators voluntarily joining and on their not coordinating to fork or abandon the chain. Fee rules themselves are changed through governance processes in which validators have voice and, historically, leverage—Section 2.1 documents validators unilaterally moving Ethereum's gas limit. The Nash product is the standard way to represent an outcome co-determined by two parties whose agreement is needed, with β indexing their relative influence; it nests the standard design benchmark, since as $\beta \to 0$ the problem converges to maximizing $\mathbb{E}[S(q)]$ subject to validators retaining nonnegative excess profits. Second, on the designer's treatment of validator costs: the validator's resource costs are not absent from the problem—they enter the bargaining objective through the profit term, with weight β, so the chosen instrument responds to cost conditions for any $\beta > 0$. Moreover, the welfare interpretation of $S(q)$ is flexible: $u(q)$ may be read as gross user surplus—Remark 2 below derives the microfounded gross-surplus specification from the valuation distribution and shows how it sharpens the comparison—or, at the cost of the closed forms, as surplus net of the resource cost of processing, in which case the instrument comparison operates through the same two channels, demand volatility and the demand–cost covariance. What the formulation deliberately allows is a designer who weights technological externalities (decentralization, propagation, state growth) that the validator does not internalize—this wedge between the two parties' valuations of block space is the source of the conflict that the fee policy must resolve. Third, on dynamics: the designer in this model commits ex-ante to a single instrument and cannot condition on the realized state. A fuller dynamic design problem would let the designer use additional levers—subsidizing users to remain in the mempool when demand is unexpectedly low, or state-contingent block rewards—but such instruments would require the protocol to observe and verify the demand state, whereas mempool data are off-chain and manipulable by the very party being regulated. The restriction to ex-ante price or quantity commitments is thus not merely an analytical simplification; it reflects the measurability constraints under which actual protocols operate, and it is the reason rule-based adjustment mechanisms such as EIP-1559, which condition only on realized block sizes, are the empirically relevant class (Section 4).

The first of these remarks can be made formal: the bargaining formulation is exactly equivalent to a standard planner's problem in which the protocol maximizes its own objective subject to a participation constraint that guarantees the validator a minimum expected excess profit, with the bargaining weight indexing the reservation level.

Lemma 2. *Fix $\beta \in (0,1)$, and let the instrument x^* (the base fee under price-setting or the block size under quantity-setting) maximize the Nash product (11), where $\Pi_x \equiv pq - C(q;\eta)$*

denotes the validator's excess profits under instrument value x, and suppose that $\mathbb{E}[S(q(x))]$ *and* $\mathbb{E}[\Pi_x]$ *are positive on the relevant choice set. Let* $\bar{\Pi}(\beta) \equiv \mathbb{E}[\Pi_{x^*}]$*. Then (i)* x^* *solves the participation-constrained design problem* $\max_x \mathbb{E}[S(q(x))]$ *subject to* $\mathbb{E}[\Pi_x] \geq \bar{\Pi}(\beta)$*; and (ii) the guaranteed profit level* $\bar{\Pi}(\beta)$ *is nondecreasing in* β*.*

Proof. Refer to Appendix C.5. □

In this sense, the Nash product is not a departure from standard mechanism design: it is a one-parameter family of participation-constrained design problems, with β playing the role of the validator's outside option. The comparative statics in β of the welfare comparison between the two regimes (Proposition 2 below) can equivalently be read as comparative statics in the rents that the protocol must concede to its validators—an object that varies observably across protocols with the cost of validator exit and the validators' role in governance. Two remarks on scope: the lemma is stated for positive objectives, such as $S(q) = q^{\nu}$, consistent with the Nash product itself; and the equivalence is within-regime—the comparison across regimes below ranks the two bargaining outcomes, under which the validator's guaranteed profit may differ across regimes, rather than holding the reservation profit fixed.

Price Controls: Assuming that the block space demand follows the isoelastic demand curve derived in (3), with a price elasticity of demand $\varepsilon > 1$ and that the block size limit is not binding, the equilibrium block size lies on the demand curve, i.e., $q = \Psi\left(\frac{p}{p_m}\right)^{-\varepsilon}$. The problem of setting optimal prices then becomes

$$\mathcal{V}^p = \max_{p \in \mathbb{R}_+} \mathbb{E}\left[S\left(\Psi\left(\frac{p}{p_m}\right)^{-\varepsilon}\right)\right]^{1-\beta} \mathbb{E}\left[(p - \Gamma) \times \Psi\left(\frac{p}{p_m}\right)^{-\varepsilon}\right]^{\beta} \tag{12}$$

Quantity Controls: If the block size is set at q, the transaction is included in the block at the price that clears markets ex-post: $p = p_m\left(\frac{q}{\Psi}\right)^{-\frac{1}{\varepsilon}}$.[17] Then, the value of setting the optimal block space is

$$\mathcal{V}^q = \max_{q \in \mathbb{R}_+} \mathbb{E}\left[S(q)\right]^{1-\beta} \mathbb{E}\left[\left(p_m\left(\frac{q}{\Psi}\right)^{-\frac{1}{\varepsilon}} - \Gamma\right) \times q\right]^{\beta} \tag{13}$$

The log-difference between the values of price controls and block space controls can be

[17] In low-demand states with $q > \Psi$, this expression extends the isoelastic demand curve below the minimum valuation p_m; under the strict microfoundation, demand is capped at Ψ and the expressions hold exactly on the scarcity event. Because the log-normal specification assigns positive probability to slack states, Proposition 2 should be read under this maintained isoelastic convention, which treats the two regimes symmetrically; the remark in Appendix C.4 discusses the convention further.

defined as follows:

$$\Delta^{\log} = \log \mathcal{V}^p - \log \mathcal{V}^q \tag{14}$$

To derive some insight into the choice between price and quantity controls, consider a blockchain designer's objective that accounts for social welfare with utility $S(q) = u(q) = q^\nu$ for $\nu > 0$. The following proposition establishes the relationship between the relative value of price controls, the price elasticity of demand, and other moments of the shock to demand and marginal costs given an arbitrary bargaining power β for the validator.

Proposition 2. *Assume that (Ψ, Γ) follows a joint log-normal distribution with log-variances $\sigma_\Psi^2, \sigma_\Gamma^2$ and log-covariance $\sigma_{\Psi,\Gamma}$.*[18] *Then, for any $\beta > 0$, the relative value of price controls over quantity controls is given by:*

$$\Delta^{\log} = \frac{1}{2}\left(\left(\hat{\nu} - \frac{\bar{\nu}}{\varepsilon}\right)\sigma_\Psi^2 - 2(\varepsilon\bar{\nu} - \beta)\sigma_{\Psi,\Gamma}\right) \tag{15}$$

where $\bar{\nu} = (1-\beta)\nu + \beta$ and $\hat{\nu} = (1-\beta)\nu^2 + \beta$.

Proof. Refer to Appendix C.6. □

We can interpret this equation first by looking at the limit for $\beta \to 1, \nu \to 1$, which gives

$$\Delta^{\log} = \frac{1}{2}(\varepsilon - 1)\left(\frac{1}{\varepsilon}\sigma_\Psi^2 - 2\sigma_{\Psi,\Gamma}\right)$$

In this case, price-setting is preferable to quantity-setting when (i) demand volatility is high and (ii) the covariance between demand and real marginal costs is low. Uncertainty in demand favors price controls, as block size adjustments can flexibly respond to demand fluctuations. Additionally, a positive correlation between marginal costs and demand disfavors price controls, as this would lead to the production of larger blocks when marginal costs are high. The price elasticity of demand, which dictates how quickly prices react to changes, mediates the degree to which the firm values (i) and (ii). A larger price elasticity of demand favors price controls. In general, these comparative statics indicate that price-setting has an advantage when demand is relatively elastic, i.e., $\varepsilon > \bar{\nu}/\hat{\nu}$, and when the covariance between demand and marginal costs is low. For $\varepsilon > 1$, the covariance coefficient $\varepsilon\bar{\nu} - \beta = \varepsilon(1-\beta)\nu + \beta(\varepsilon - 1)$ is positive, so a positive covariance always weighs against price controls. For $\nu \in (1 - 1/\varepsilon, 1)$, the weight on demand volatility rises with the validator's

[18] Stating the assumption on the marginal cost Γ itself is the natural primitive form: for cost functions that are homogeneous of degree one with a multiplicative shock, such as the Cobb–Douglas example of Equation (9), $\Gamma(\eta, g_x)$ is proportional to η, so joint log-normality of (Ψ, η) and of (Ψ, Γ) coincide.

bargaining power β while the covariance penalty declines with it, so that higher validator bargaining power tilts the comparison toward price controls.

Remark 1. *The joint log-normality of* (Ψ, Γ) *should be read as a tractable benchmark rather than a substantive restriction. The expression for* $\Delta^{\log}$ *depends on the distributions only through terms of the form* $\log \mathbb{E}[\Psi^a \Gamma^b]$*, i.e., through the cumulant-generating function of* $(\log \Psi, \log \Gamma)$*. For arbitrary distributions with small shocks, a second-order cumulant expansion of these terms yields exactly Equation (15), with* σ_Ψ^2 *and* $\sigma_{\Psi,\Gamma}$ *reinterpreted as the variance and covariance of log-deviations; higher-order cumulants (skewness, kurtosis) contribute terms that vanish faster than* σ^2*. This is the same sense in which Weitzman's (1974) original comparison is a local quadratic approximation. Log-normality makes the formula exact and globally valid within the maintained isoelastic convention, which matters for the calibration in Section 4, where fee and demand fluctuations are large; heavy-tailed multiplicative specifications of this kind are standard descriptions of blockchain fee data (Guo et al., 2019).*

Remark 2. *The designer's objective* $S(q) = q^\nu$ *can be tied directly to the valuation microfoundation. Under the Pareto distribution of Lemma 1, the gross surplus of the included users in state* Ψ *at block size* q *equals* $\frac{\varepsilon}{\varepsilon-1} p_m \Psi^{1/\varepsilon} q^{(\varepsilon-1)/\varepsilon}$*: an isoelastic function of block space with curvature* $\nu = (\varepsilon - 1)/\varepsilon$*, scaled by a state multiplier* $\Psi^{1/\varepsilon}$ *that raises the value of space in high-demand states. The specification* $S(q) = q^\nu$ *thus captures the quantity dimension of microfounded surplus—nesting it exactly at* $\nu = (\varepsilon - 1)/\varepsilon$*—while abstracting from the state multiplier. Reinstating the multiplier leaves the optimal instruments unchanged, since it varies with neither* p *nor* q*, and adds* $(1-\beta)\frac{\nu}{\varepsilon}\sigma_\Psi^2 > 0$ *to* $\Delta^{\log}$*: a designer who values block space more in high-demand states finds the quantity flexibility of price controls correspondingly more attractive. (At this boundary value* $\nu = (\varepsilon-1)/\varepsilon$*, the covariance coefficient* $\varepsilon\bar{\nu} - \beta$ *equals* $\varepsilon - 1$ *and is independent of* β*.) Appendix C.6 presents the calculation. More broadly,* $S(q)$ *should be read as designer preferences over the quantity dimension—spanning technological targets and surplus-based objectives—rather than as exact expected surplus.*

Taking Stock: The above result helps us understand the differences in the policies that determine fees and block space in Bitcoin and Ethereum. We can see that both blockchains are instances where price elasticity of demand is high $\varepsilon > 1$ and demand uncertainty is significant $\sigma_\Psi^2 \gg 0$.

The background provided in Section 2.1 suggests that Ethereum can be viewed through the lens of this model as an instance where the bargaining power of the validator is high $\beta \to 1$ (or is considered high by the protocol designers). Thus, our use of 'high bargaining power' for validators should be interpreted in the sense of their economic incentive alignment in the transaction-fee market, rather than an overriding governance control of the protocol overall. In addition, since the proof-of-stake upgrade, the marginal cost for Ethereum validators has been virtually constant, so $\sigma_{\Psi,\Gamma} \approx 0$. Proposition 2 states that, in this case, price-setting is the most favorable policy. This explains the adoption of the EIP-1559 price-setting policy for the Ethereum blockchain. Furthermore, Figure 1 shows that EIP-1559 has performed better since Ethereum's proof-of-stake upgrade, consistent with the correlation between the validator's marginal costs and demand being lower relative to miners' in a proof-of-work protocol.

Bitcoin is distinguished as the first blockchain with an anonymous founder, and its core developers strongly emphasize decentralization and censorship resistance. As discussed in Section 2.1, most proposals to change Bitcoin's fees and block size policies have failed. Bitcoin can, therefore, be viewed through the lens of our model as an instance where the bargaining power of validators (here miners) is low $\beta \to 0$ (or is not attributed enough importance by protocol designers, given the low block space utilization rate still observed). In addition, under the proof-of-work protocol, the cost of marginal block space—driven primarily by orphan risk, which is proportional to the value at stake in the block, and secondarily by the per-transaction component of energy costs, which varies with the season (Borenstein and Holland, 2005) and co-moves with cryptocurrency market cycles (Rehman and Kang, 2021)—is largely positively correlated with demand, $\sigma_{\Psi,\Gamma} \gg 0$. Proposition 2 states that, in this case, price-setting is less effective.

Although the model treats β as a primitive, its assignment across protocols is not free: it is disciplined by observable governance outcomes, and only the ranking matters for the comparison. On Ethereum, block producers repeatedly and unilaterally moved the gas limit before EIP-1559 (Appendix A documents this history), and again in 2025, when validator signaling doubled the limit from 30 to 60 million without a hard fork; core development is concentrated—Fracassi et al. (2024) find that ten individuals proposed 68% of implemented core EIPs—and validators capture fees and MEV directly. On Bitcoin, the corresponding miner-supported initiatives to change capacity (notably the big-block proposals culminating in the 2017 fork) failed against developer and user opposition. Translating such governance records into a cardinal β is beyond this paper's scope; by Lemma 2, the comparative statics require only the ordinal statement that block producers have had more influence over Ethereum's supply parameters than over Bitcoin's. Two further clarifications guard against over-reading. First, β parameterizes whose payoffs the chosen rules must respect—by Lemma 2, the rents conceded to validators—not who drafts the rules; the adoption of EIP-1559, which curbed validators' interim discretion while formalizing their compensation through tips and value capture, is consistent with a designer optimizing subject to substantial validator rents. Second, the gas-limit episodes evidence validators' influence over supply parameters—the variable bargained over in the model—rather than over the choice of regime itself, which the designer makes subject to those rents.

In general, the key determinants of the advantage of price-setting in blockchains are the validator's bargaining power, the elasticity of demand, the validator's uncertainty about demand, and the covariance of demand and marginal costs. This insight will guide us in Section 4 in studying the implications of my results for Ethereum transaction fees.

3.5 Effect of Cryptocurrency Price Fluctuations

Cryptocurrency prices can be volatile, while users value transaction processing services in dollars or real terms. The choice of the dollar (equivalently, the consumption good) as numeraire reflects the economics of both sides of this market: validators' input costs—electricity, hardware, bandwidth—are invoiced in fiat, and the outside options of most users (centralized exchanges, traditional payment rails) are denominated in fiat, so willingness to pay for transaction inclusion is most naturally measured in real terms. The protocol's instruments, by contrast, are necessarily expressed in the native currency: a base fee is quoted in gwei, not in dollars. This mismatch between the unit in which the price control is posted and the unit in which valuations and costs are denominated is exactly what this section captures. This section examines the impact of cryptocurrency price volatility on the choice between price and quantity controls. Let P denote the exchange rate between 1 USD and the cryptocurrency (equivalently, the inverse of the cryptocurrency price expressed in dollars). Another way to interpret P is the exchange rate between 1 unit of consumption goods and the cryptocurrency. This real model accommodates variations in cryptocurrency price and fiat currency's value. Users pay transaction fees in the cryptocurrency at a nominal price p, implying that the dollar (equiv. real) value of these payments is $\frac{p}{P}$. Meanwhile, Γ represents the dollar (equiv. real) marginal cost.

The following proposition, formulated for simplicity with $\beta = 1$ (though similar insights apply for other parameters), provides an equivalent to Proposition 2 in this context:

Proposition 3. *Suppose (Ψ, Γ, P) is jointly log-normal distributed. Then, the relative value of price adjustments over quantity adjustments is*

$$\Delta^{\log} = \frac{1}{2}(\varepsilon - 1)\left(\frac{1}{\varepsilon}\sigma_\Psi^2 - 2\sigma_{\Psi,\Gamma} - \varepsilon\sigma_P^2 - 2\varepsilon\sigma_{P,\Gamma}\right) \tag{16}$$

Proof. Refer to Appendix C.7. □

In addition to the findings of Proposition 2, the variance of the cryptocurrency price and the covariance between the cryptocurrency price and the dollar (equiv. real) marginal cost both decrease the relative advantage of price controls over quantity controls.

3.6 Which Assumptions Are Essential?

It is useful to take stock of the model's key assumptions and distinguish those that are essential for the results from those that are simplifications of convenience.

Interim Monopoly of the Validator. This assumption is essential. The conflict between the designer and the validator—and hence the entire instrument-choice problem—exists because the consensus mechanism grants the block proposer discretion over block contents. With perfectly competitive block production at the interim stage, fees would equal marginal cost and the choice of instrument would be moot.

Ex-Ante Commitment to a Single Instrument. This assumption is essential, and it is institutionally grounded. Fee rules are encoded in the protocol and can condition only on on-chain observables. State-contingent instruments would require verifiable measures of demand that validators cannot manipulate, and no such measures exist because mempool data are off-chain. This is the blockchain counterpart of the assumption in Weitzman (1974) that the regulator moves before uncertainty resolves.

Nash Bargaining over Instruments. The bargaining formulation is essential for the comparative static in the validator's bargaining power, and it is relaxable at the extremes. By Lemma 2, the formulation is equivalent to a standard design problem subject to a validator participation constraint, with β indexing the guaranteed rents. The polar cases $\beta \to 0$ and $\beta \to 1$ reproduce, respectively, a pure designer problem subject to validator participation and a pure monopolist problem; the bargaining formulation interpolates between them and maps governance differences across protocols into a single parameter. The qualitative roles of demand uncertainty, of the covariance between costs and demand, and of the demand elasticity are present for any fixed $\beta > 0$.

Linear Validator Costs. The per-unit cost structure is a simplification, and what matters is its variable component. Costs that depend on block contents discipline the validator's supply decision, while consensus-layer costs are fixed with respect to block composition and affect only participation, as discussed in Section 3.1. A strictly convex cost of block space would preserve the Weitzman-style logic, with the cost curvature additionally entering the comparison.

Truthful Bidding. Truthful bidding is exact under price controls and a robust approximation under quantity controls, as established in Section 3.2 and Proposition 1.

Isoelastic Demand from Pareto Valuations. This specification is a simplification with empirical support. Lemma 1 maps any valuation distribution into an elasticity; the Pareto case makes the elasticity constant and the formulas closed-form, and heavy tails are a documented feature of fee data (Guo et al., 2019). The optimal-adjustment result of Section 4 is local and requires only the elasticity at the target. Remark 2 ties the designer objective $S(q) = q^{\nu}$ to the surplus generated under the Pareto microfoundation.

Joint Log-Normality of Shocks. Log-normality is a benchmark that delivers exact and globally valid formulas. By Remark 1, the welfare comparison coincides with a second-order cumulant expansion for general distributions with small shocks.

One-Period Mempool Exit. The exit assumption is a simplification. Queueing makes demand for next-block inclusion more elastic, and this margin is absorbed into the definition and measurement of the elasticity, as discussed in Sections 3.2 and 4.

4 The Dynamics of Fee Policies and Implications for Ethereum

In this section, I explore how my results can inform the design of Ethereum fee policies. Under the bargaining objective with $S(q) = q^{\nu}$ and moment conditions in (15) favoring price-setting, Proposition 2 implies that committing to prices ex-ante and letting quantities adjust dominates committing to a fixed block size. For a designer who cares only about a technological target, Claim 2 cuts the other way: a quantity commitment attains the target exactly. Ethereum's fee policies thread these two benchmarks by pursuing the target with a price instrument: as discussed in Section 2, the base fee adjusts adaptively so that blocks average the target size q^{target} while quantities remain free to absorb demand within each block. How can prices be set iteratively and heuristically? Below, I define a family of simple TFMs, including EIP-1559, Ethereum's TFM. I study their dynamics, determine the shape and adjustment rate of the optimal mechanism within this family, and provide bounds on the target block size that align with the incentives of a monopolistic validator.

Definition 1. *(ADT-TFM) A TFM is called adaptive to a deterministic target (ADT) if there exists a deterministic block size, q^{target} (the target), and a deterministic function, g (the adjustment function), such that the base fee satisfies*

$$\frac{p_{t+1}}{p_t} = g\left(\frac{q_t - q^{target}}{q^{target}}\right) \tag{17}$$

Example 3. *(EIP-1559) The base fee in EIP-1559 is ADT with linear adjustment function $g(x) = 1 + d \times x$ where the adjustment parameter is $d = \frac{1}{8}$.*

Let q^* denote the optimal quantity control or the block size that a blockchain designer aims to achieve for a specific technological target: S penalizes deviations from q^{target} as in Example 2, in the limiting case placing a Dirac mass at the target, $S(q) = \delta\{q - q^{target}\}$. In this case, $q^* = q^{target}$. We consider a general demand curve, with price elasticity of demand $\varepsilon(q^{target})$ that is assumed to be fixed and uncertainty in demand represented by λ. It is worth being explicit about the benchmark against which optimality is evaluated in this section. The designer's objective here is the technological block-size target itself: deviations of realized block sizes from q^{target} are the welfare loss, as in the squared-loss objective of Claim 2, because persistent deviations strain propagation, state growth, and node requirements, while transient deviations correspond to misallocated block space and avoidable fee volatility for users. A TFM is then considered *robustly optimal* if, following a sudden change or shock in demand, it brings the realized quantity as close as possible to the targeted level in the worst-case demand scenario—a minimax criterion over demand realizations. This is a deliberately conservative benchmark: it does not require the protocol to know the distribution of demand shocks, only the price elasticity of demand at the target, in keeping with the informational constraints under which actual fee rules operate. When I conclude below that Ethereum's base fee adjusts "faster than optimal," the statement is relative to this benchmark: an adjustment parameter above the inverse elasticity systematically overshoots the price change needed to return block sizes to target after a demand shock, generating excess oscillation in both block sizes and fees rather than faster convergence. The following proposition determines the shape and slope of the optimal ADT-TFM.

Proposition 4. *Suppose that the demand curve is log-convex; the robustly optimal ADT-TFM has an adjustment parameter equal to the inverse price elasticity of demand, $d = \frac{1}{\varepsilon(q^{target})}$, and exponential form: $g(x) = \exp(d \cdot x)$ and*

$$p_{t+1} = p_t \exp\left(\frac{1}{\varepsilon(q^{target})} \frac{q_t - q^{target}}{q^{target}}\right) \tag{18}$$

The intuition is transparent in a local linearization. Let x_t denote the percentage deviation of the realized block size from the target. Along a demand curve with elasticity $\varepsilon(q^{target})$, undoing a deviation x_t requires a log price adjustment of $x_t/\varepsilon(q^{target})$: when demand is highly elastic, a small price correction moves a large quantity, so the price should react little; when demand is inelastic, prices must move aggressively to restore the target. An ADT-TFM changes log prices by $\ln g(x_t) \approx d\,x_t$. Setting $d = 1/\varepsilon(q^{target})$ therefore corrects a demand shift in a single step, and the exponential form makes the prescription exact in log deviations—$\ln g(\xi) = \xi/\varepsilon$ globally—when the elasticity is constant. The formal proof in Appendix C.8 establishes that this choice is also robustly optimal when the size of the demand shift is unknown: any steeper rule overshoots the target in the worst case, and any flatter rule undershoots it.

If we restrict the adjustment function to be linear, as in EIP-1559, or of the form $(1+d)^{\frac{q_t-q^{target}}{q^{target}}}$ as studied by Leonardos et al. (2023), the optimal adjustment rate remains the inverse of the price elasticity of demand, since these rules share the same local slope. This elasticity, however, captures the price elasticity of inclusion of a transaction in the next block. Given the user interface shown in Figure 2, users are less likely to change their price in a short period, so this elasticity will be larger than the price elasticity of inclusion of a transaction within 5–10 minutes, for which users can substitute waiting in the mempool.

The Dynamics of Fee Adjustment: Proposition 4 is a worst-case, one-shot statement: it selects the rule that best corrects a single demand shift of unknown size. The same parameter restriction also governs the behavior of the fee mechanism away from the worst case. The following result characterizes the local dynamics of block sizes and base fees under any ADT-TFM and shows that the inverse elasticity is simultaneously the point of fastest stable adjustment, the minimizer of stationary fee and block-size volatility, and the onset of oscillatory behavior. The random-walk specification of demand complements the cross-sectional log-normal specification of Section 3: the instrument-choice problem concerns the distribution of demand at the commitment horizon, while the adjustment problem concerns its evolution block by block.

Proposition 5 (Local fee dynamics)**.** *Consider an ADT-TFM with adjustment function satisfying $g(x) = 1 + dx + o(x)$ (this includes EIP-1559's linear rule and the exponential rule of Proposition 4), an isoelastic demand curve with elasticity ε, and log demand following a random walk, $\ln \Psi_{t+1} = \ln \Psi_t + w_{t+1}$, with i.i.d. innovations of variance σ_w^2. Let x_t denote the log deviation of the block size from the target. Then, to first order:*

(i) Block-size deviations follow the AR(1) process $x_{t+1} = (1-d\varepsilon)\,x_t + w_{t+1}$, with persistence

$1 - d\varepsilon$.

(ii) Deviations decay monotonically if $0 < d \leq 1/\varepsilon$, *oscillate with damping if* $1/\varepsilon < d < 2/\varepsilon$, *and admit no stationary distribution (local instability) if* $d \geq 2/\varepsilon$.

(iii) The stationary variance of block-size deviations, $\sigma_w^2/\left(d\varepsilon(2-d\varepsilon)\right)$*—and that of the base fee's deviation from the market-clearing price, which equals it divided by* ε^2*—is uniquely minimized at* $d = 1/\varepsilon$.

(iv) The variance of base-fee changes, $d\,\sigma_w^2/\left(\varepsilon(2-d\varepsilon)\right)$, *is strictly increasing in* d.

Proof. Refer to Appendix C.9. □

Two remarks delimit the scope of this characterization. The law of motion in (i) requires neither the isoelastic nor the random-walk specification: the derivation is a local linearization, so it holds for any demand curve with local elasticity $\varepsilon(q^{target})$ at the target, and for any log-demand process, with the innovation w_{t+1} equal to realized demand growth. The isoelastic and random-walk specifications are used only for the stationary-variance formulas in (iii)–(iv) and to make the linearization globally accurate.

Proposition 5 sharpens the welfare assessment of adjustment speed in three ways. First, it shows that $d = 1/\varepsilon$ is not merely robust to the worst case (Proposition 4) but variance-minimizing in the stochastic steady state: faster adjustment does not buy faster convergence to target—it buys overshooting, which raises the volatility of both block sizes and fees. Second, it locates two thresholds with policy bite: oscillations begin exactly at $d = 1/\varepsilon$, and local instability sets in at $d = 2/\varepsilon$. Third, it provides a theoretical account of existing simulation evidence: Leonardos et al. (2021) and Leonardos et al. (2023) find stable behavior of the (nonlinear) EIP-1559 dynamics only for adjustment rates up to roughly 8%, with chaotic behavior for larger rates. Through the lens of Proposition 5, 8% is approximately the oscillation threshold $1/\varepsilon$ at the elasticity I estimate below, and the chaotic regimes correspond to the oscillatory and locally unstable regions of the linearized system.

Adjustment Parameter for Ethereum: I now approximate Ethereum's adjustment rate through the lens of my analysis. I use Ethereum data because of its widespread availability, the simplicity of its ADT-TFM, and the global usage of its blockchain. A parallel calibration for Bitcoin is not undertaken, and the asymmetry should be acknowledged: under a first-price regime without a posted reserve, observed fees embed equilibrium bid shading, so the tail of paid fees does not identify the valuation tail the way post–EIP-1559 data do. The assignment of a positive cost–demand covariance to Bitcoin therefore rests on the orphan-risk and energy-price channels discussed in the introduction and the studies cited there (Rehman and Kang, 2021; Delgado-Mohatar et al., 2019), not on a structural estimate; I flag this as a limitation of the comparative exercise.

A random sample of 100,000 blocks, encompassing 16,881,386 transactions, was extracted from the complete set of Ethereum blocks.[19] This sample spans from the introduction of EIP-1559 at the London hard fork (block number 12965000, August 5, 2021) to block number 17731768 (July 20, 2023). Additional random block subsamples from before and after the Ethereum merge (block number 15537393, September 15, 2022) are also analyzed.[20]

The median block in the sample contains 143 transactions. Each block is associated with a number and a timestamp, the total gas used by all transactions in the block (equivalent to q in the model), and an array of transactions. Each transaction includes information on its gas unit, gas price, and other metadata. I do not have random variation in supply to trace the demand curve here. Instead, informed by Lemma 1, I calculate the Pareto tail of transaction gas prices—as the nearest proxy for user valuations—for blocks of size around q^{target} to obtain the price elasticity $\varepsilon(q^{target})$. The tail coefficient is estimated within each block by maximum likelihood and then averaged across the blocks of the indicated subset; Appendix D details the estimator and the aggregation. This approach has limitations that I discuss in Appendix D.

My preferred estimate is a Pareto coefficient of 12.45, obtained for blocks of size within $\pm 5\%$ of the block size target with a maximum base fee of 200 gwei (7,189 blocks; see Table A2), which yields an optimal adjustment rate of 8.03%. This is significantly below Ethereum's current adjustment rate of 12.5%. This result aligns with the finding of Leonardos et al. (2023), who simulate the dynamic system of EIP-1559 and find stability around the target block size only for adjustment parameters below 8%. Several alternative choices in Appendix D yield a range between 6.14% and 11% for the Ethereum optimal adjustment rate. My contribution clarifies that the adjustment rate encapsulates the economic concept of inverse price elasticity of demand, which must be measured or approximated on-chain.

The policy interpretation of this calibration deserves emphasis. First, the finding does not say that EIP-1559 was a mistake—on the contrary, Proposition 2 rationalizes Ethereum's move to price controls—but that a single parameter of the mechanism, chosen at the time as a round number rather than estimated, is set too high. The cost of overshooting is concrete: after a demand shock, a base fee that moves by 12.5% per block when the elasticity warrants 8% repeatedly overcorrects, producing oscillations of block sizes around the target and excess base-fee volatility, which is costly for users precisely along the predictability

[19] The data are drawn entirely from the public Ethereum blockchain. Appendix D describes the sampling frame, variables, estimator, and aggregation in sufficient detail for the analysis to be reproduced from any archival node or public blockchain dataset.

[20] The results are consistent when I separately sample 100,000 blocks from before and after the merge.

dimension that EIP-1559 was designed to improve (Liu et al., 2022). In terms of Proposition 5, every elasticity estimate in Tables A2–A4 places the current rate $d = 12.5\%$ above the oscillation threshold $1/\varepsilon$ (which ranges from 6.1% to 11% across estimates), and the largest tail estimate ($\varepsilon = 16.29$, in the pre-merge sample) places it above the local instability threshold $2/\varepsilon \approx 12.3\%$. Second, the analysis supplies the missing economic interpretation of the parameter: d is not a free tuning constant but the inverse of the price elasticity of demand for next-block inclusion, an object that can be estimated from on-chain data and re-estimated as the user base changes. Third, the estimates come with the caveats of Appendix D: censoring of low valuations by the base fee pushes the implied d up, while the mechanical compression of effective gas prices from above pushes it down, so the point estimate is best read as a tentative benchmark rather than a precise parameter; what is robust—taking the estimator at face value—is that every estimate across the grid of sample restrictions lies below the current 12.5%. These caveats counsel implementing changes through the same conservative governance process as other parameter changes, not a mechanical re-fitting.

Target Block Size and MEV: Now, let us determine the block size target that aligns with the optimal target of a monopolistic validator. The goal of such a block size target is to make the blockchain immune to a simple form of MEV—that is, to prevent the validator from including her value-extracting transactions while reducing the effective supply available to users. The following definition makes this notion explicit.

Definition 2. *(Myopic-Miner Incentive Compatibility) A quantity target is myopic-miner incentive-compatible (MMIC) if a myopic miner, by creating no fake transactions and adhering to the suggested block size target q^{target}, maximizes her profit.*

The MMIC definition implies that a miner who aims to maximize her revenue should be motivated to comply with the proposed quantity target when choosing her block size ex-ante.

Proposition 6. *Given any isoelastic demand curve, myopic-miner incentive compatibility requires the target block size to satisfy*

$$\frac{q^{target}}{q^{\max}} \leq \left(\frac{\varepsilon}{\varepsilon - 1}\right)^{-\varepsilon} \mathbb{E}\left[\lambda^{\frac{1}{\varepsilon}}\right]^{\varepsilon}, \tag{19}$$

with equality in the binding case $p_m = \mathbb{E}[\Gamma]$, in which the minimum user valuation equals the expected marginal cost.

Proof. Refer to Appendix C.10. □

This proposition states that the maximum block size should have an adequate buffer above the block size target; otherwise, if the block size target is too high relative to the maximum block size, the monopolist validator will have incentives to fill the block up to the target size with her value-extracting transactions. Two distinct deviations discipline the target, and it is worth separating them. In the fee-retention regime analyzed in the proof (quantity controls, e.g., Bitcoin and pre–EIP-1559 Ethereum), the first deviation is over- or under-supplying *real* transactions: the target is self-enforcing only if it coincides with the validator's monopoly optimum over the user demand curve, which is what generates the formula. The second deviation is filling residual capacity with the validator's *own* transactions: because fee payments on self-included transactions are a wash under retention, the marginal unit costs only $\mathbb{E}[\Gamma]$, while the validator's own value-extracting transactions are worth at least the minimum user valuation p_m per unit (the normalization adopted in the proof), so deterring self-inclusion requires $p_m \leq \mathbb{E}[\Gamma]$—the condition that turns the monopoly optimum into the upper bound of the proposition. Under EIP-1559's base-fee burning, the second deviation changes character: stuffing destroys the base fee on every self-included transaction, which is exactly the discipline that burning is designed to provide (Roughgarden, 2020), and the binding concern becomes MEV-motivated stuffing—self-included transactions whose private value exceeds the burned base fee plus marginal cost. The level of the burned base fee then takes over the deterrent role that $p_m \leq \mathbb{E}[\Gamma]$ plays under retention: because the base fee that sustains a given target is decreasing in the target, a lower target raises the per-unit cost of MEV-motivated stuffing. A full characterization of MMIC targets under burning, replacing the minimum user valuation with the validator's private per-unit value of inclusion, is left for future work. In particular, if $q^{\max}$ is adjusted to coincide with user demand under average demand conditions, then $\frac{q^{\max}}{q^{target}} > e$. For the price elasticity found above, the fee-retention benchmark implies that the ratio of the maximum block size to the block size target should be greater than 2.84, against Ethereum's current ratio of 2; under base-fee burning, this comparison should be read through the MEV channel discussed above—a lower target raises the burned base fee that an MEV-motivated validator must pay—rather than as a direct critique of Ethereum's parameters.

A note on market structure is also warranted. Since the merge, much of Ethereum block construction operates under proposer–builder separation (PBS), in which specialized builders assemble blocks and proposers (validators) auction off the right to determine contents. PBS relocates rather than eliminates the interim monopoly on which the model rests: for a given slot, a single builder ultimately holds discretion over the block's composition, and competition among builders shifts the division of MEV between builders and proposers without restoring price-taking behavior at the block-composition stage. The model's "validator" should accordingly be read as the proposer–builder coalition that controls contents in a slot; Bahrani et al. (2024) analyze transaction fee mechanism design when such MEV-driven intermediaries are modeled explicitly.

5 Design Guidance and Managerial Implications

The analysis yields operational guidance for protocol designers and decentralized autonomous organization (DAO) governance bodies, together with implications for managers of congestible platforms beyond blockchains. I summarize it here.

Choosing the instrument. The choice between a price control and a quantity control should be made from four measurable data: the volatility of demand for block space; the covariance between validators' marginal costs and demand, which is governed by the consensus technology (positive under proof of work, approximately zero under proof of stake); the price elasticity of demand, recoverable from the tail of the on-chain fee distribution (Lemma 1); and the rents that the protocol must concede to its validators, which Lemma 2 maps to the bargaining weight and which vary observably with validators' role in governance. Proposition 2 then prescribes price controls when demand volatility is high, the cost–demand covariance is low, demand is elastic, and validator rents are substantial (the bargaining-power comparative static applies for the empirically relevant curvature $\nu \in (1 - 1/\varepsilon, 1)$)—and quantity controls in the converse configuration. Every input to this decision rule is estimable from chain data or observable protocol characteristics.

Tuning the adjustment rate. For protocols on the price-control regime, the base-fee adjustment rate is not a free tuning constant: it should equal the inverse of the price elasticity of demand for next-block inclusion (Propositions 4 and 5), estimated from the Pareto tail of fees in blocks near the target and re-estimated as the composition of demand changes. For Ethereum, the estimates point toward lowering the adjustment rate from 12.5% toward roughly 8% (range 6.1%–11% across the estimates in Appendix D, and subject to the measurement caveats discussed there). The expected gains are concrete: less overshooting after demand shocks, lower base-fee volatility, and dynamics at the boundary of the monotone region rather than well inside the oscillatory one (Proposition 5). Because congestion-driven fee spikes disproportionately price out smaller users (Cong et al., 2023), calibrating the adjustment rate is also an inclusion instrument.

Setting the target-to-maximum ratio. In regimes where fee revenue accrues to the validator—quantity-control protocols such as Bitcoin, and any design without base-fee burning—deterring validators from filling residual capacity with their own value-extracting transactions requires a block-size target no greater than approximately 37% of the block size that meets demand in the average state, and roughly 35% at the measured elasticity (Proposition 6). Under base-fee burning, the fee-recycling attack is unprofitable for myopic validators, and the deterrent against MEV-motivated stuffing is the level of the burned base fee itself, which a lower target raises. The model also clarifies how to read the gas-limit doubling that validators enacted in 2025: because EIP-1559 moves the target one-for-one with the limit, the target-to-capacity ratio remained unchanged at one-half; what disciplines the validator is the level of the target relative to demand—the bound in Proposition 6 does not involve the capacity itself—so capacity policy and target policy should be assessed jointly.

Beyond blockchains. The framework applies to any platform that allocates congestible capacity through rule-based pricing while the short-run allocation decision rests with an agent holding temporary market power. The classical service-pricing literature characterizes optimal prices for congestible systems under a single operator (Naor, 1969; Mendelson, 1985; Afeche and Mendelson, 2004). The present analysis adds the two ingredients that recur on decentralized platforms: the designer cannot coerce the operator, so instruments must respect the operator's rents (Lemma 2); and rules can condition only on realized, verifiable quantities, so prices must adjust through elasticity-calibrated feedback rules rather than state-contingent schedules.

For regulators. As crypto-assets underlie exchange-traded products, the predictability of transaction settlement depends on the stability properties of fee mechanisms. Proposition 5 supplies supervisors with a concrete, estimable statistic—the product of the adjustment rate and the demand elasticity—that determines whether a blockchain's fee rule operates in its monotone, oscillatory, or unstable region.

6 Conclusion

In this paper, I have studied different blockchain fee policies to allocate block space efficiently when various sources of uncertainty exist. Namely, I have compared the quantity-setting regime used by Bitcoin and the price-setting regime used by Ethereum under different sources of uncertainty. Price controls are optimal in an environment with high validator bargaining power, high price elasticity of demand, high demand uncertainty, and a low correlation between validators' marginal costs and demand; quantity controls are favored when marginal costs co-move with demand and validators' bargaining power is limited. In addition, a high variance of the cryptocurrency price mitigates the advantages of price controls. These insights are robust to strategic user bidding: truthful bidding is an equilibrium property of the price-control mechanisms, and under quantity controls, equilibrium bids of atomistic users coincide with the market-clearing price in the scarcity states in which fee policy binds, leaving the welfare comparison unchanged. These insights help explain the difference between Bitcoin's and Ethereum's fee policies. Next, I have applied these insights to a family of fee policies to which the method used by Ethereum belongs. Under mild assumptions on the shape of the demand for block space, I find that a crucial parameter of the mechanism, the rate of adjustment of prices, equals the inverse price elasticity of demand for inclusion in the next block. Some calculations using Ethereum transaction data suggest that the rate at which Ethereum's fees change is faster than optimal. Beyond the level of the adjustment rate, I characterize the dynamics it induces: block-size deviations decay with persistence equal to one minus the adjustment-rate–elasticity product, oscillations begin exactly at the inverse elasticity, and the same value minimizes stationary fee volatility—a theoretical counterpart to the stability thresholds found in simulations of EIP-1559.

Henceforth, understanding the economics of multidimensional fees is crucial, as transactions use various types of resources. The question of designing fee policies for durable resources is growing in importance as blockchain states expand significantly over time, particularly due to resources such as storage. Additionally, as some blockchain states may experience less congestion than others, it is imperative to explore fee policies that differentiate pricing based on varying demand levels rather than solely pricing block space.[21] While a comprehensive examination of these complex issues lies beyond the scope of this paper, they offer promising opportunities for future research and further refinement of my analysis.

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Appendix

A History of Block Size Changes in Ethereum

Dates	Changes in Gas Limit	Reasons and Context
March 4th, 2016	3,141,592 to 4,712,388	Max gas increased by 1.5x, and minimum gas price reduced from 50 to 20 gwei (a denomination of the Ethereum cryptocurrency) to improve affordability and throughput.
September 22nd, 2016	4,712,388 to 1,000,000	Network experienced a DDoS attack, leading to a drastic reduction in block size for security measures.
September 22nd, 2016	1,000,000 to 1,500,000	DDoS attack was partially mitigated, allowing a limited increase in block size.
October 15th, 2016	1,500,000 to 500,000	Block size was decreased as a precautionary step before implementation of a hard fork to address security vulnerabilities.
October 19th, 2016	500,000 to 3,000,000	Hard fork successfully implemented, security issues resolved, leading to a substantial increase in block size.
October 20th, 2016	3,000,000 to 1,500,000	Another DDoS attack occurred, prompting a reduction in block size to safeguard the network.
October 23rd, 2016	1,500,000 to 2,000,000	Successful mitigation of attacks led to increased block size, signaling a return to stability.
November 24th, 2016	2,000,000 to 3,300,000	Continued stability and growing user base justified another increase in block size.
December 5th, 2016	3,300,000 to 4,000,000	Network achieved consistent stability, allowing a further increase in block size.
June 3rd, 2017	4,000,000 to 4,712,388	Increase in network activity and user engagement necessitated a rise in block size.

June 29th, 2017	4,712,388 to 6,283,184	New target gas limit set at 4,700,000 to better match growing ecosystem demands.
December 10th, 2017	6,283,184 to 8,000,000	Rise in transaction volume driven by Cryptokitties NFT required an increase in block size.
September 19th, 2019	8,000,000 to 10,000,000	Tether's migration to the Ethereum blockchain led to increased transaction demands, prompting a block size increase.
June 19th, 2020	10,000,000 to 12,000,000	Miner consensus to increase block size was reached, supporting the growing Ethereum ecosystem.
July 25th, 2020	12,000,000 to 12,500,000	Minor increase following another round of miner agreements aimed at fine-tuning network performance.
April 21st, 2021	12,500,000 to 15,000,000	Berlin hard fork led to efficiency improvements, enabling a substantial increase in block size.
August 5th, 2021	15,000,000 to 30,000,000	London hard fork brought about major improvements in transaction fee predictability and network efficiency, justifying a block size doubling.

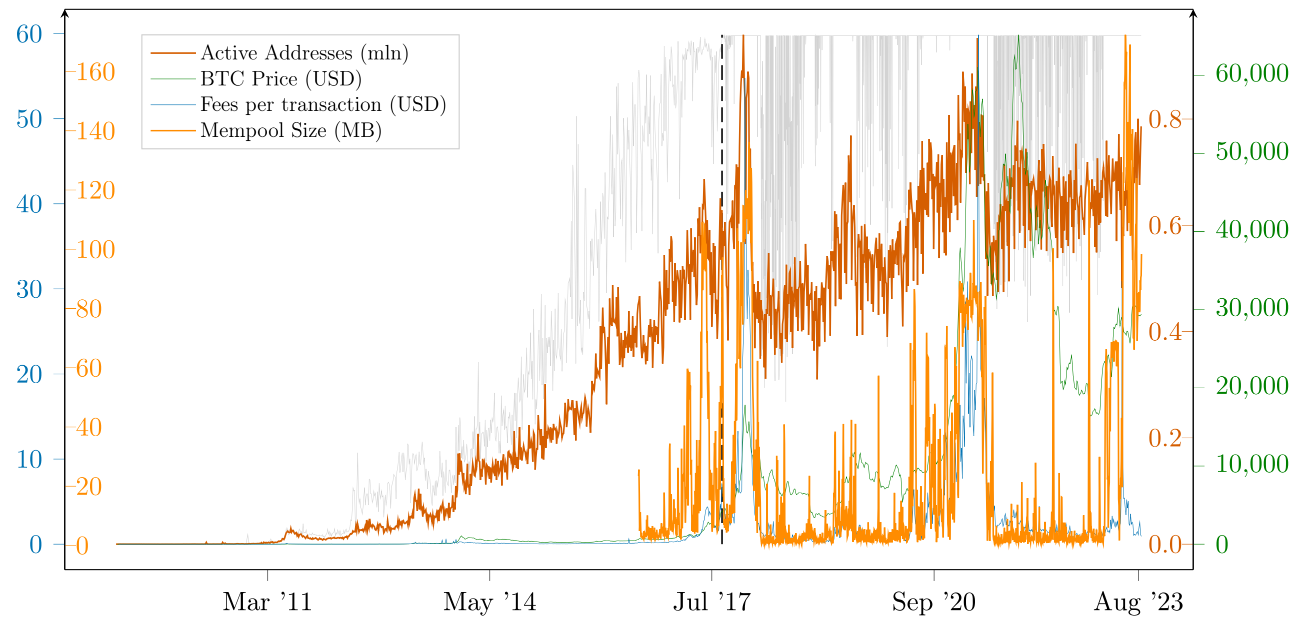
Table A1: Table of history

B Capacity Utilization

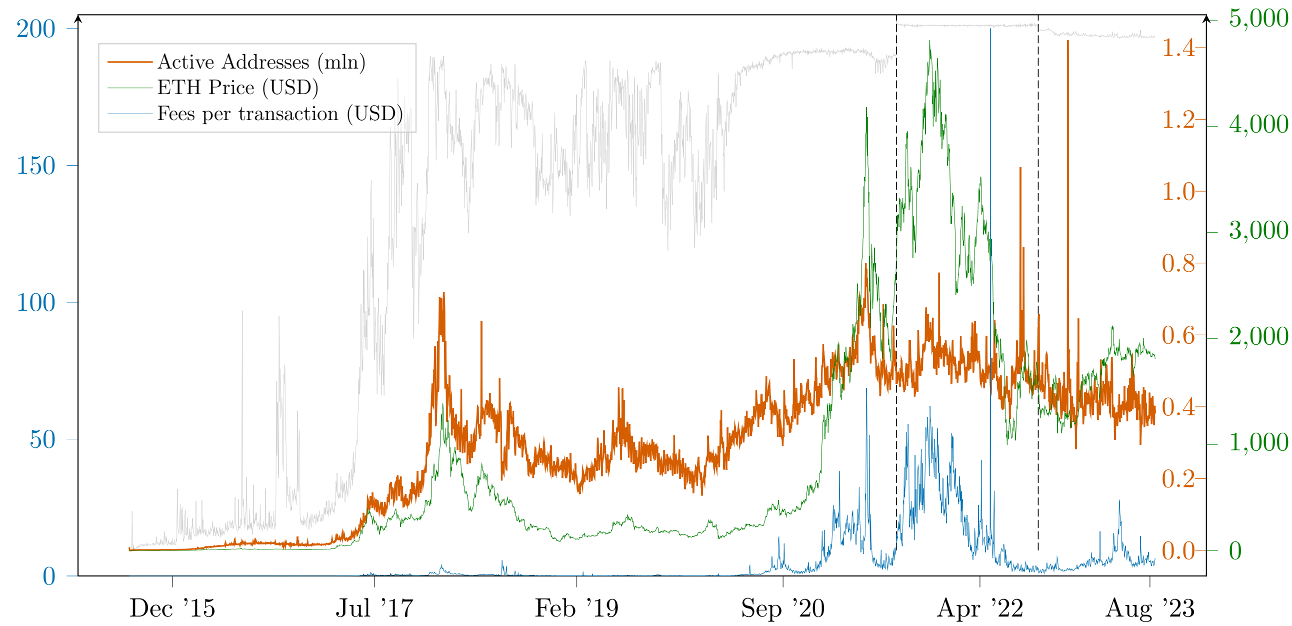
B.1 Demand Factors

For my analysis, I examine exogenous demand shifters—the number of active addresses and token prices—together with transaction fees and, for Bitcoin (BTC) only, the size of the mempool, i.e., the pool of transactions that has yet to be confirmed and included in a block. The latter two variables are endogenous congestion outcomes that proxy for demand conditions rather than shift them.

For BTC, the top graph of Figure A1 shows that the number of active addresses has a strong correlation with the block utilization rate, as it acts as a good proxy for demand. Similarly, transaction fees show a tight link with the utilization rate, capturing the fact that users must offer a higher fee to be picked by miners in periods of block congestion. As explained in Lehar and Parlour (2020), mempool size is not a perfect predictor of block utilization, as miners can leave blocks empty, even when there are transactions waiting, to extract more profits. Finally, the BTC price has a loose relationship with the utilization



(a) Bitcoin network demand factors per day against utilization rate



(b) Ethereum network demand factors per day against utilization rate

Figure A1: Time series of Bitcoin and Ethereum demand factors rate, possibly because of its high volatility due to speculation.

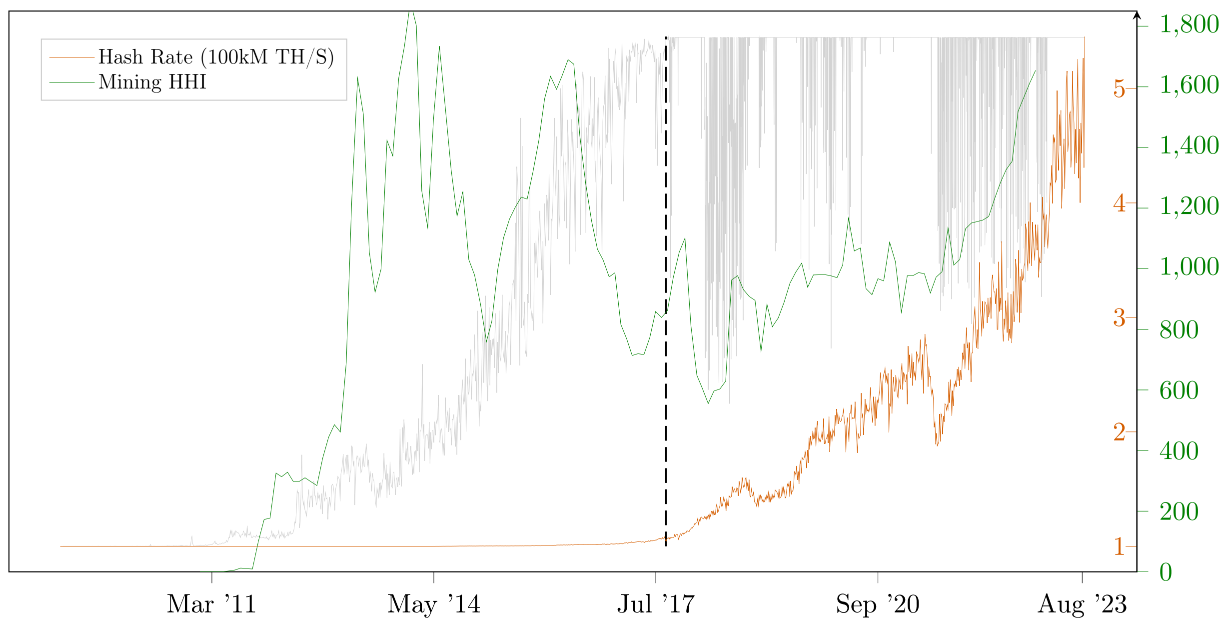
For Ethereum (ETH), the bottom graph of Figure A1 shows that, before the London fork, the number of active addresses once again had the highest correlation with block utilization rates, representing best users' demand for transactions. Both transaction fees and the ETH price have stronger correlations with block fullness than their BTC counterparts. After the implementation of EIP-1559, gas levels stabilized consistently at the target level, suggesting that the implementation of the new fee system successfully achieved its objective. As a consequence, the block utilization rate became uncorrelated with the demand-side variables that previously tracked it. As noted in Section 2.1, this stabilization is the mechanical implication of the base-fee rule operating as designed, and the post–EIP-1559 period coincides with Layer-2 rollup adoption and large swings in cryptocurrency markets; the decoupling is therefore evidence on the mechanism's operation rather than a causal estimate of EIP-1559's effect on demand.

B.2 Supply Factors

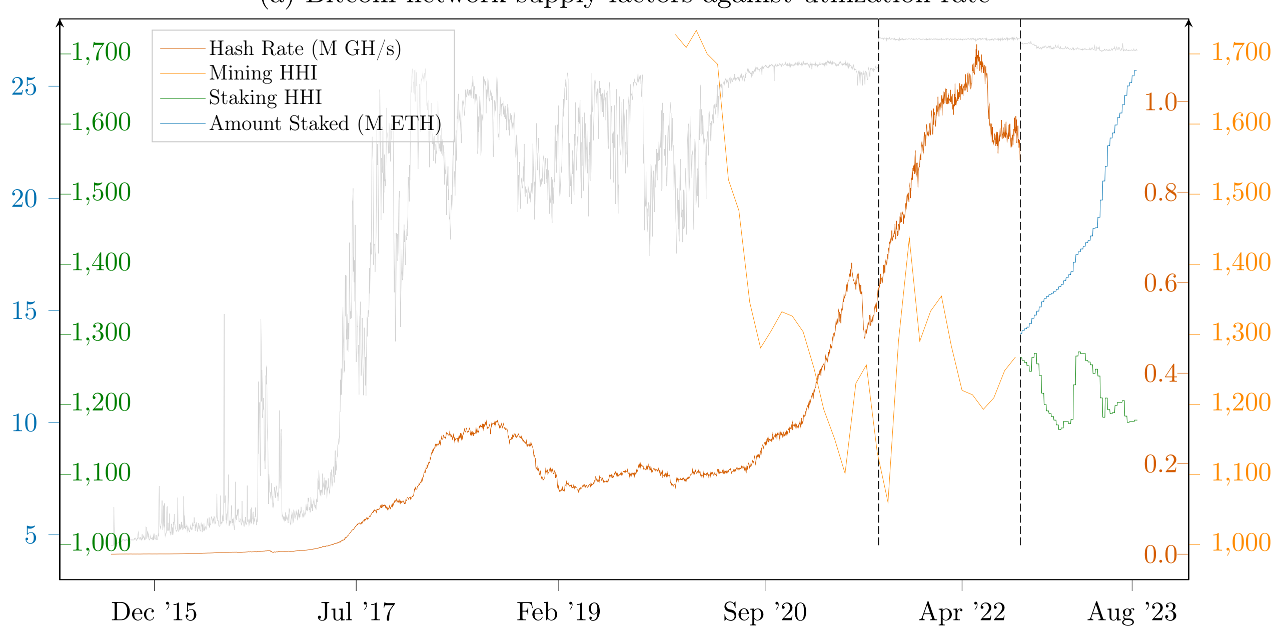
For my analysis, I identified the computing power of miners and the concentration rate of mining pools as potential determinants of block space supply. For ETH only, I considered the total amount of ETH staked and the concentration of validator pools after the proof-of-stake update.

For BTC, the top graph of Figure A2 shows no discernible association between the amount of computing power invested into mining and the block utilization rate. Both series exhibit strong secular growth over the sample, so these plots should be read as descriptive; establishing or ruling out a causal relationship would require a formal time-series analysis that is beyond the scope of this appendix. I measure it using the hash rate, which measures how many guesses are made per second to solve the code to mine the next block. Regarding mining pool concentration, computed as the Herfindahl–Hirschman index (the sum of the squares of individual market shares) from the shares of blocks mined in a one-month period, the graph seems to support the hypothesis that miners exercise their market power by leaving blocks less than full, as periods of higher HHI are usually accompanied by utilization rates below 100%.

For ETH, the bottom graph of Figure A2 shows that, under both proof of work and proof of stake, the computing power (or the amount of ETH staked, of which 32 ETH are required to activate a validator software) is not visibly associated with the utilization rate, similarly to what we observe for BTC. In contrast to the BTC case, however, the pool concentration is not associated with the block utilization rate either. The HHI measures (from the share of



(a) Bitcoin network supply factors against utilization rate



(b) Ethereum network supply factors against utilization rate

Figure A2: Time series of Bitcoin and Ethereum supply factors blocks mined in a one-month period under proof of work and from the share of ETH staked over a one-month period under proof of stake) are not correlated with the utilization rate, which is particularly noticeable after the stabilization of the utilization rate post–London fork.

C Analytic Proofs

C.1 Proof of Lemma 1

At a price p, demand for block space is the measure of users' willingness to pay p for transaction inclusion, i.e., $\lambda q^{\max}\bar{F}(p)$. This yields the demand curve $p = \left(\bar{F}\right)^{-1}\left(\frac{q}{\lambda q^{\max}}\right)$. The price elasticity of demand is defined by (negative) the percentage change in quantity demanded over the percentage change in price, i.e., $-\frac{dq/q}{dp/p}$. Since from the demand curve $q = \lambda q^{\max}\bar{F}(p)$ and $dq/dp = -\lambda q^{\max} f(p)$, the demand elasticity is $\frac{pf(p)}{1-F(p)}$.

When F is a Pareto distribution with scale p_m and shape α,

$$\bar{F}(p) = Pr(v > p) = \begin{cases} \left(\frac{p_m}{p}\right)^{\alpha} & \text{for } p \geq p_m \\ 1 & \text{for } p < p_m \end{cases} \tag{20}$$

so that, above the minimum price p_m, demand is

$$q = \lambda q^{\max}\left(\frac{p_m}{p}\right)^{\alpha}$$

Therefore, we obtain

$$\frac{p}{p_m} = \left(\frac{q}{\lambda q^{\max}}\right)^{-\frac{1}{\alpha}}$$

C.2 Proof of Claim 1

Proof. Available next-block demand at prices at or below the minimum valuation p_m equals Ψ, and at prices $p \geq p_m$ it equals $\Psi(p_m/p)^{\varepsilon}$; no price creates transactions beyond the arrivals Ψ. The quantity control $q^{\max}$ therefore delivers $q = \min\{\Psi, q^{\max}\}$, the largest feasible block, in every state. Under a price control p, realized size is $q = \min\{\Psi(p_m/p)^{\varepsilon}, q^{\max}\}$, which falls strictly short of $\min\{\Psi, q^{\max}\}$ whenever $p > p_m$ and the cap does not bind—a deadweight loss in low-demand states relative to the quantity control, which establishes (i). For (ii), demand at price $p \geq p_m$ is $\Psi(p_m/p)^{\varepsilon} < q^{\max}$ whenever $p > p_m\lambda^{1/\varepsilon}$; along the isoelastic extension of the demand curve below p_m (the convention discussed in Section 3.4), $q = q^{\max}$ exactly at $p = p_m\lambda^{1/\varepsilon}$, and $p_m\lambda^{1/\varepsilon} \to 0$ as $\lambda \to 0$. ∎

C.3 Proof of Claim 2

The optimal ex-ante quantity choice is just $q = q^{target}$, for which the loss is zero (maintaining the convention that demand suffices to reach the target in every state). The optimal ex-ante price choice solves

$$\min_{p \in \mathbb{R}^+} \mathbb{E}\left[\left(\Psi \cdot \left(\frac{p}{p_m}\right)^{-\varepsilon} - q^{target}\right)^2\right] \tag{21}$$

Denote $x \equiv \left(\frac{p}{p_m}\right)^{-\varepsilon}$. The first-order condition for the choice of p (resp. x) in (21) is

$$x = \frac{\mathbb{E}[\Psi]q^{target}}{\mathbb{E}[\Psi^2]} \tag{22}$$

The expected block size is then

$$\mathbb{E}[\Psi x] = \frac{\mathbb{E}[\Psi]^2 q^{target}}{\mathbb{E}[\Psi^2]} \leq q^{target} \tag{23}$$

Replacing (22) in the value of the loss function (21) yields

$$\frac{\mathbb{E}[\Psi^2] - \mathbb{E}[\Psi]^2}{\mathbb{E}[\Psi^2]}(q^{target})^2 = \frac{\mathrm{Var}(\Psi)}{\mathbb{E}[\Psi^2]}(q^{target})^2$$

C.4 Proof of Proposition 1

Fix the realized demand state $\Psi = \lambda q^{\max}$ and the block size q, with $q < \Psi$ so that block space is scarce. Let $p^* = p_m(q/\Psi)^{-1/\varepsilon}$ denote the market-clearing price, defined by $\Psi\bar{F}(p^*) = q$: the measure of users with valuations of at least p^* exactly equals the available block space.

Consider the strategy profile in which every user with $v_j \geq p^*$ bids $b_j = p^*$ and every user with $v_j < p^*$ abstains. Under this profile, the measure of submitted transactions is $\Psi\bar{F}(p^*) = q$, all bids are identical, and the validator includes all of them, filling the block exactly. We verify that no user has a profitable deviation; because users are atomistic, no individual deviation changes the clearing price, the set of competing bids, or the validator's inclusion rule.

- A user with $v_j \geq p^*$ who deviates to a bid $b < p^*$ is outranked by a measure q of bids equal to p^* and is excluded, forfeiting a nonnegative surplus $v_j - p^* \geq 0$.
- A user with $v_j \geq p^*$ who deviates to a bid $b > p^*$ remains included but, under pay-as-bid pricing, pays $b > p^*$, strictly reducing her surplus.

- A user with $v_j < p^*$ who deviates to any bid $b \geq p^*$ is included with probability at most one and pays at least $p^* > v_j$, earning negative surplus; bids below p^* leave her excluded, as in the candidate profile.

Hence, the profile is a Nash equilibrium. In this equilibrium, the included transactions are exactly those of users with $v_j \geq p^*$—the same set as in the market-clearing benchmark, where the clearing price allocates the block to the q highest valuations—the realized block size is q, and the validator's revenue is $p^* q$, the value used in the quantity-control objective (13). (Note the benchmark prices all included transactions at the clearing price; a hypothetical pay-as-bid auction with truthful bids would instead extract inframarginal surplus and yield revenue above $p^* q$.) Since $\mathcal{V}^q$ depends on the bidding game only through the realized quantity and fee revenue, the quantity-control outcome coincides with the market-clearing benchmark state by state on the scarcity event, and therefore $\mathcal{V}^q$ and the regime comparison of Proposition 2 are unchanged. ∎

Remark. Four comments on the scope of this construction. First, the statement is in the continuum (large-market) limit of the Poisson arrival process: with finitely many bidders, each bidder is pivotal with positive probability, and the equivalence holds asymptotically in the sense of Swinkels (2001). Second, the construction treats the marginal user as indifferent and uses the continuum to break ties: at the profile above, the block is exactly filled, so rationing does not occur on a positive measure of users. Third, in slack states ($q \geq \Psi$), competition for inclusion disappears and pay-as-bid revenue falls below the benchmark formula, which extends the isoelastic demand curve below the Pareto scale p_m; the benchmark of Equation (13) is itself an approximation in such states, and the comparison of the two regimes should be read as driven by the scarcity states in which the fee policy binds. Fourth, when users observe Ψ with noise, bids must forecast the clearing price and the exact equivalence becomes approximate; Swinkels (2001) establishes asymptotic efficiency for equilibria of large pay-as-bid (and uniform-price) multi-unit auctions, which also mitigates the dependence of the conclusion on equilibrium selection.

C.5 Proof of Lemma 2

Proof. (i) Suppose some x satisfied $\mathbb{E}[\Pi_x] \geq \bar{\Pi}(\beta)$ and $\mathbb{E}[S(q(x))] > \mathbb{E}[S(q(x^*))]$. Then x would attain a strictly higher value of $\mathbb{E}[S]^{1-\beta}\,\mathbb{E}[\Pi]^{\beta}$ than x^*, contradicting the optimality of x^*. (ii) Write the log-objective as $f(x;\beta) = \log \mathbb{E}[S(q(x))] + \beta\, B(x)$, where $B(x) \equiv \log \mathbb{E}[\Pi_x] - \log \mathbb{E}[S(q(x))]$. Let $\beta_2 > \beta_1$, with respective maximizers x_2, x_1. Adding the two optimality inequalities $f(x_1;\beta_1) \geq f(x_2;\beta_1)$ and $f(x_2;\beta_2) \geq f(x_1;\beta_2)$ yields $(\beta_2 -$

$\beta_1)(B(x_2) - B(x_1)) \geq 0$, hence $\log \mathbb{E}[\Pi_{x_2}] - \log \mathbb{E}[\Pi_{x_1}] \geq \log \mathbb{E}[S(q(x_2))] - \log \mathbb{E}[S(q(x_1))]$. Suppose, for contradiction, that $\mathbb{E}[\Pi_{x_2}] < \mathbb{E}[\Pi_{x_1}]$. The displayed inequality then forces $\mathbb{E}[S(q(x_2))] < \mathbb{E}[S(q(x_1))]$, so x_1 strictly dominates x_2 in both factors of the objective, and $f(x_2; \beta_2) < f(x_1; \beta_2)$, contradicting the optimality of x_2 at β_2. □

C.6 Proof of Proposition 2

We first find the expression of the marginal cost Γ in (10). The first-order condition is

$$c_i(x_1, \ldots, x_N; \eta) = \gamma g_{x_i} \tag{24}$$

where γ is the Lagrangian of constraint (2). Since c is homogeneous of degree 1, we have

$$c(x_1, \ldots, x_N; \eta) = \sum_{i=1}^{N} c_i(x_1, \ldots, x_N; \eta) x_i = \gamma \sum_{i=1}^{N} g_{x_i} x_i = \gamma q \tag{25}$$

Therefore, $\Gamma = \gamma$. Evaluating at $q = 1$ yields

$$\Gamma(\eta, g_x) = c(x_1(1, g_x, \eta), \ldots, x_N(1, g_x, \eta); \eta) \tag{26}$$

To prove the proposition, we now take logarithms of the blockchain designer's objective and maximize over p and q. The first-order conditions for the price choice and quantity choice are

$$\frac{\varepsilon(1-\beta)\nu}{p^*} + \frac{\varepsilon\beta}{p^*} = \frac{\beta \mathbb{E}[\Psi]}{p^* \mathbb{E}[\Psi] - \mathbb{E}[\Gamma\Psi]} \tag{27}$$

$$\frac{(1-\beta)\nu}{q^*} + \frac{\beta}{q^*} = \frac{\beta/\varepsilon p_m (q^*)^{-1-1/\varepsilon} \mathbb{E}[\Psi^{1/\varepsilon}]}{p_m (q^*)^{-1/\varepsilon} \mathbb{E}[\Psi^{1/\varepsilon}] - \mathbb{E}[\Gamma]} \tag{28}$$

Denote $\bar{\nu} = (1-\beta)\nu + \beta$. Then, we obtain

$$p^* = \frac{\varepsilon\bar{\nu}}{\varepsilon\bar{\nu} - \beta} \frac{\mathbb{E}[\Gamma\Psi]}{\mathbb{E}[\Psi]} \tag{29}$$

$$q^* = \left(\frac{\varepsilon\bar{\nu}}{\varepsilon\bar{\nu} - \beta} \frac{1}{p_m} \frac{\mathbb{E}[\Gamma]}{\mathbb{E}[\Psi^{1/\varepsilon}]} \right)^{-\varepsilon} \tag{30}$$

The log values of price controls and quantity controls are then

$$\log \mathcal{V}^p = -\varepsilon\bar{\nu} \log(p^*/p_m) + (1-\beta)\log(\mathbb{E}[\Psi^\nu]) + \beta \log(p^* \mathbb{E}[\Psi] - \mathbb{E}[\Psi\Gamma]) \tag{31}$$

$$\log \mathcal{V}^q = \bar{\nu} \log(q^*) + \beta \log(p_m (q^*)^{-1/\varepsilon} \mathbb{E}[\Psi^{1/\varepsilon}] - \mathbb{E}[\Gamma]) \tag{32}$$

Replacing the optimal choices with their values in (29) and (30), we obtain

$$\begin{aligned} \log \mathcal{V}^p &= -\varepsilon\bar{\nu}\log(p^*/p_m) + (1-\beta)\log(\mathbb{E}[\Psi^\nu]) + \beta\log(\frac{\beta}{\varepsilon\bar{\nu}-\beta}\mathbb{E}[\Psi\Gamma]) && (33)\\ \log \mathcal{V}^q &= \bar{\nu}\log(q^*) + \beta\log(\frac{\beta}{\varepsilon\bar{\nu}-\beta}\mathbb{E}[\Gamma]) && (34)\end{aligned}$$

Simplifying yields

$$\begin{aligned} \log \mathcal{V}^p - \log \mathcal{V}^q &= \bar{\nu}\varepsilon\log\mathbb{E}[\Psi] && (35)\\ &+(\bar{\nu}\varepsilon-\beta)(\log\mathbb{E}[\Gamma]-\log\mathbb{E}[\Psi\Gamma])\\ &-\varepsilon\bar{\nu}\log\mathbb{E}[\Psi^{1/\varepsilon}] + (1-\beta)\log\mathbb{E}[\Psi^\nu]\end{aligned}$$

For the joint log-(Ψ, Γ), with mean

$$\mu = \begin{pmatrix}\mu_\Psi\\ \mu_\Gamma\end{pmatrix}$$

and variance–covariance matrix

$$\Sigma = \begin{pmatrix}\sigma_\Psi^2 & \sigma_{\Psi,\Gamma}\\ \sigma_{\Psi,\Gamma} & \sigma_\Gamma^2\end{pmatrix}$$

we have

$$\begin{aligned} \log\mathbb{E}[\Psi] &= \mu_\Psi + \frac{1}{2}\sigma_\Psi^2 && (36)\\ \log\mathbb{E}[\Gamma]-\log\mathbb{E}[\Psi\Gamma] &= -\mu_\Psi - \frac{1}{2}\sigma_\Psi^2 - \sigma_{\Psi,\Gamma} && (37)\\ \log\mathbb{E}[\Psi^{1/\varepsilon}] &= \frac{1}{\varepsilon}\mu_\Psi + (\frac{1}{\varepsilon^2})\frac{1}{2}\sigma_\Psi^2 && (38)\\ \log\mathbb{E}[\Psi^\nu] &= \nu(\mu_\Psi + \frac{1}{2}\nu\sigma_\Psi^2) && (39)\end{aligned}$$

Putting them together, the terms in μ_Ψ cancel, and we obtain the result

$$\log \mathcal{V}^p - \log \mathcal{V}^q = \frac{1}{2}\left(\left(\hat{\nu} - \frac{\bar{\nu}}{\varepsilon}\right)\sigma_\Psi^2 - 2(\varepsilon\bar{\nu}-\beta)\sigma_{\Psi,\Gamma}\right) \quad (40)$$

where $\hat{\nu} = (1-\beta)\nu^2 + \beta$.

Microfounded welfare (Remark 2). Under the Pareto distribution, the gross surplus of included users in state Ψ at block size $q \le \Psi$ is

$$\Psi \int_{p(q)}^{\infty} v\, dF(v) = \Psi \frac{\varepsilon}{\varepsilon-1} p_m^{\varepsilon} p(q)^{1-\varepsilon} = \frac{\varepsilon}{\varepsilon-1} p_m \Psi^{1/\varepsilon} q^{(\varepsilon-1)/\varepsilon},$$

using $p(q) = p_m (q/\Psi)^{-1/\varepsilon}$. Replacing $\mathbb{E}[S(q)]$ by $\mathbb{E}[\Psi^{1/\varepsilon} S(q)]$ multiplies the designer factor of each objective by terms that do not vary with the instrument, so the first-order conditions and the optima (29)–(30) are unchanged. The designer terms in the log values become $(1-\beta)\log\mathbb{E}[\Psi^{\nu+1/\varepsilon}]$ under price controls (where $q = \Psi(p/p_m)^{-\varepsilon}$) and $(1-\beta)\log\mathbb{E}[\Psi^{1/\varepsilon}] + (1-\beta)\nu\log q^*$ under quantity controls. Using $\log\mathbb{E}[\Psi^a] = a\mu_\Psi + \frac{1}{2}a^2\sigma_\Psi^2$, the comparison shifts by

$$(1-\beta)\left[\tfrac{1}{2}\left(\nu+\tfrac{1}{\varepsilon}\right)^2 - \tfrac{1}{2}\nu^2 - \tfrac{1}{2}\tfrac{1}{\varepsilon^2}\right]\sigma_\Psi^2 = (1-\beta)\,\frac{\nu}{\varepsilon}\,\sigma_\Psi^2,$$

as stated in Remark 2.

C.7 Proof of Proposition 3

Users value transaction processing in real terms, so a nominal price p corresponds to the real price p/P, and demand is $q = \Psi\left(\frac{p}{Pp_m}\right)^{-\varepsilon}$, while the real marginal cost is Γ. With $\beta = 1$, the value of committing to a nominal price is

$$\mathcal{V}^p = \max_p\ p^{-\varepsilon} p_m^{\varepsilon}\left(p\,\mathbb{E}[\Psi P^{\varepsilon-1}] - \mathbb{E}[\Gamma\Psi P^{\varepsilon}]\right),$$

whose monopoly first-order condition gives $p^* = \frac{\varepsilon}{\varepsilon-1}\mathbb{E}[\Gamma\Psi P^{\varepsilon}]/\mathbb{E}[\Psi P^{\varepsilon-1}]$ and hence, up to constants common to both regimes,

$$\log\mathcal{V}^p = \varepsilon\log\mathbb{E}[\Psi P^{\varepsilon-1}] - (\varepsilon-1)\log\mathbb{E}[\Gamma\Psi P^{\varepsilon}] + \text{const}.$$

Under a quantity commitment, the nominal clearing price is $p = Pp_m(q/\Psi)^{-1/\varepsilon}$, so the real revenue per unit, p/P, does not involve P, and $\log\mathcal{V}^q = \varepsilon\log\mathbb{E}[\Psi^{1/\varepsilon}] - (\varepsilon-1)\log\mathbb{E}[\Gamma] + \text{const}$, exactly as in the proof of Proposition 2 at $\beta = 1$. For jointly log-normal (Ψ, Γ, P), expanding each $\log\mathbb{E}[\cdot]$ with $\log\mathbb{E}[X^aY^b] = a\mu_X + b\mu_Y + \frac{1}{2}(a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY})$, the mean terms cancel, the $\sigma_{\Psi,P}$ terms cancel ($\varepsilon(\varepsilon-1) - (\varepsilon-1)\varepsilon = 0$), and collecting the rest yields

$$\Delta^{\log} = \tfrac{1}{2}\left(\varepsilon - (\varepsilon-1) - \tfrac{1}{\varepsilon}\right)\sigma_\Psi^2 - (\varepsilon-1)\sigma_{\Psi,\Gamma} + \tfrac{1}{2}\left(\varepsilon(\varepsilon-1)^2 - (\varepsilon-1)\varepsilon^2\right)\sigma_P^2 - (\varepsilon-1)\varepsilon\,\sigma_{\Gamma,P},$$

which simplifies to $\frac{1}{2}(\varepsilon-1)\left(\frac{1}{\varepsilon}\sigma_\Psi^2 - 2\sigma_{\Psi,\Gamma} - \varepsilon\sigma_P^2 - 2\varepsilon\sigma_{P,\Gamma}\right)$, as stated. ■

C.8 Proof of Proposition 4

Setup and notation. Demand in block t satisfies $q_t = \lambda_t q^{\max} \bar{F}(p_t)$, where λ_t is the unknown demand intensity. Write the base-fee update of an ADT-TFM as $\ln p_{t+1} = \ln p_t + \ln g((q_t - q^{target})/q^{target})$, and let $\xi_t \equiv \ln(q_t/q^{target})$ denote the log deviation from the target; to first order, the percentage deviation $(q_t - q^{target})/q^{target}$ equals ξ_t, and we work with ξ_t. Define log inverse demand as a function of the log of its argument, $h(y) \equiv \ln \bar{F}^{-1}(e^y)$, so that along the demand curve $\ln p_t = h(\ln q_t - \ln(\lambda_t q^{\max}))$, with derivative $h' = -1/\varepsilon(\cdot)$ evaluated at the corresponding quantity, where ε is the demand elasticity of Lemma 1.

Step 1: the required adjustment. The worst case for a rule that must correct using a single observation is a persistent shift: suppose the intensity moves to λ_t and remains there at $t+1$. The price that restores the target in the next block satisfies

$$\ln p^c_{t+1} - \ln p_t \;=\; h\big(\ln q^{target} - \ln(\lambda_t q^{\max})\big) - h(\ln q_t - \ln(\lambda_t q^{\max}))\,,$$

a difference of h between two points whose arguments differ by exactly $-\xi_t$, at a common but unknown location indexed by λ_t.

Step 2: constant elasticity. If the elasticity is constant, h is affine with slope $-1/\varepsilon$, so the required adjustment equals ξ_t/ε for every realization of λ_t. The rule $\ln g(\xi) = \xi/\varepsilon$—the exponential adjustment function with rate $d = 1/\varepsilon$, expressed in log deviations—therefore returns the block size exactly to target in one step after every persistent demand shift, attaining the minimax value of zero deviation; under the percentage-deviation convention of the ADT definition, the same rule corrects persistent shifts up to second order in the deviation. It is the unique such rule: any rule with $\ln g(\xi) \neq \xi/\varepsilon$ for some admissible deviation ξ fails to correct the persistent shift that produces that deviation, leaving a worst-case deviation bounded away from zero (at first order, under the percentage-deviation convention).

Step 3: general demand. For a general demand curve, the mean value theorem gives the required adjustment as $\xi_t/\varepsilon(\tilde{q})$ for some quantity $\tilde{q}$ between q^{target} and q_t along the realized demand curve. An ADT rule conditions only on ξ_t, not on λ_t or on the location of $\tilde{q}$. For small deviations, a rule with local slope a at zero induces the error $(a - 1/\varepsilon(q^{target}))\,\xi_t + O(\xi_t^2)$ against the persistent-shift benchmark, so avoiding first-order worst-case errors requires $a = 1/\varepsilon(q^{target})$, i.e., an adjustment rate equal to the inverse elasticity at the target. When the demand curve is log-convex, so that h is convex and the local elasticity varies monotonically along the curve, this local prescription is unambiguous (recall that the elasticity at the target,

$\varepsilon(q^{target})$, is held fixed across demand states, as assumed in the text). The characterization is thus exact under constant elasticity and holds to first order in the deviation from target for general log-convex demand, with the exponential form the canonical member of the first-order-optimal class—exact whenever the elasticity is constant near the target. ■

C.9 Proof of Proposition 5

Let $x_t = \ln q_t - \ln q^{target}$. With isoelastic demand, $q_t = \Psi_t(p_t/p_m)^{-\varepsilon}$, so that

$$x_t = \ln \Psi_t - \varepsilon \ln p_t + \varepsilon \ln p_m - \ln q^{target}.$$

The ADT-TFM updates the base fee by

$$\ln p_{t+1} = \ln p_t + \ln g\left(\frac{q_t - q^{target}}{q^{target}}\right) = \ln p_t + d\, x_t + o(x_t),$$

using $g(x) = 1 + dx + o(x)$ and $(q_t - q^{target})/q^{target} = e^{x_t} - 1 = x_t + o(x_t)$. Substituting,

$$x_{t+1} = x_t + (\ln \Psi_{t+1} - \ln \Psi_t) - \varepsilon\,(\ln p_{t+1} - \ln p_t) = (1 - d\varepsilon)\, x_t + w_{t+1} + o(x_t),$$

which is claim (i). Claim (ii) reads off the autoregressive root $\rho = 1 - d\varepsilon$: deviations decay monotonically when $\rho \in [0, 1)$, i.e., $0 < d\varepsilon \leq 1$; alternate in sign with damping when $\rho \in (-1, 0)$, i.e., $1 < d\varepsilon < 2$; and admit no stationary distribution when $|\rho| \geq 1$, i.e., $d\varepsilon \geq 2$. These statements concern the linearized dynamics; they are valid in the small-noise limit in which deviations remain in a neighborhood of the target.

For claim (iii), with $|\rho| < 1$ the stationary variance of the AR(1) is

$$\mathrm{Var}(x) = \frac{\sigma_w^2}{1 - \rho^2} = \frac{\sigma_w^2}{d\varepsilon\,(2 - d\varepsilon)}.$$

The denominator $d\varepsilon(2 - d\varepsilon)$ is strictly concave in $d\varepsilon$ and uniquely maximized at $d\varepsilon = 1$, so the variance is uniquely minimized at $d = 1/\varepsilon$. For the base fee, the market-clearing price at the target satisfies $\ln p_t^c = (\ln \Psi_t - \ln q^{target})\,/\varepsilon + \ln p_m$, so the fee's deviation from the clearing level is $u_t \equiv \ln p_t - \ln p_t^c = -x_t/\varepsilon$ identically; hence $\mathrm{Var}(u) = \mathrm{Var}(x)/\varepsilon^2$, minimized at the same point.

For claim (iv), base-fee changes satisfy $\ln p_{t+1} - \ln p_t = d\, x_t + o(x_t)$, so their variance is

$$d^2\,\mathrm{Var}(x) = \frac{d^2\,\sigma_w^2}{d\varepsilon(2 - d\varepsilon)} = \frac{d\,\sigma_w^2}{\varepsilon\,(2 - d\varepsilon)},$$

which is strictly increasing in d on $(0, 2/\varepsilon)$. ■

C.10 Proof of Proposition 6

From equation (30), the optimal quantity for a monopolistic miner is, for $\beta = 1$,

$$q^* \;=\; \left(\frac{\varepsilon}{\varepsilon - 1}\frac{1}{p_m}\frac{\mathbb{E}[\Gamma]}{\mathbb{E}[\Psi^{1/\varepsilon}]}\right)^{-\varepsilon} \tag{41}$$

With $\Psi = \lambda q^{\max}$. Now, suppose the minimum user valuation exceeded the expected marginal cost, $p_m > \mathbb{E}[\Gamma]$. Adherence to the target is optimal against deviations in real supply only if the target coincides with this monopoly optimum, $q^{target} = q^*$. Consider next the self-inclusion deviation, and assume that the validator's own value-extracting transactions are worth at least the minimum user valuation p_m per unit of block space—the natural normalization for transactions whose purpose is to extract value from the chain. Under fee retention, fee payments on self-included transactions are a wash—the validator pays them to herself—so filling a unit of residual capacity costs her only the expected marginal processing cost $\mathbb{E}[\Gamma]$ and yields at least p_m. A myopic validator facing $p_m > \mathbb{E}[\Gamma]$ would therefore fill blocks to the limit rather than adhere to an interior target, breaking MMIC. Hence MMIC requires $p_m \leq \mathbb{E}[\Gamma]$; thus,

$$q^* \;\leq\; q^{\max}\left(\frac{\varepsilon - 1}{\varepsilon}\right)^{\varepsilon}\mathbb{E}[\lambda^{1/\varepsilon}]^{\varepsilon} \tag{42}$$

By Jensen's inequality, $\mathbb{E}[\lambda^{1/\varepsilon}]^{\varepsilon} \leq \mathbb{E}[\lambda]$. Thus, the block size limit is set to match user demand in expectation; then, $\mathbb{E}[\lambda] = 1$, and thus,

$$\frac{q^*}{q^{\max}} \;\leq\; \left(\frac{\varepsilon - 1}{\varepsilon}\right)^{\varepsilon} \tag{43}$$

The right-hand side is an increasing function for $\varepsilon > 1$ with limit e^{-1}.

These bounds provide valuable insights for studies of fee policies involving a monopolistic validator (Nisan, 2023; Lavi et al., 2022) and fee policies designed to prevent the validator from monopolizing all the surplus (Bahrani et al., 2024).

Figure A3 illustrates how the upper bound of the ratio $q^*/q^{\max}$ changes with ε. This upper bound approaches an asymptote of $e^{-1} \approx 37\%$, and it reaches 95% of this limit when $\varepsilon = 10.42$. Ethereum currently sets the target block size to half the block size limit. The correct interpretation of this result is that any ADT-TFM in which fee revenue accrues to the validator and whose target block size exceeds 37% of the block size that meets user demand in an average demand scenario would not be invulnerable to a simple form of MEV; under base-fee burning, the fee-recycling channel is shut for myopic validators, and the discussion after Proposition 6 delineates how the burned base fee takes over the deterrent role. This is because MMIC necessitates that the validator has no incentive to include her own value-extracting transactions while simultaneously reducing the effective supply available to users.

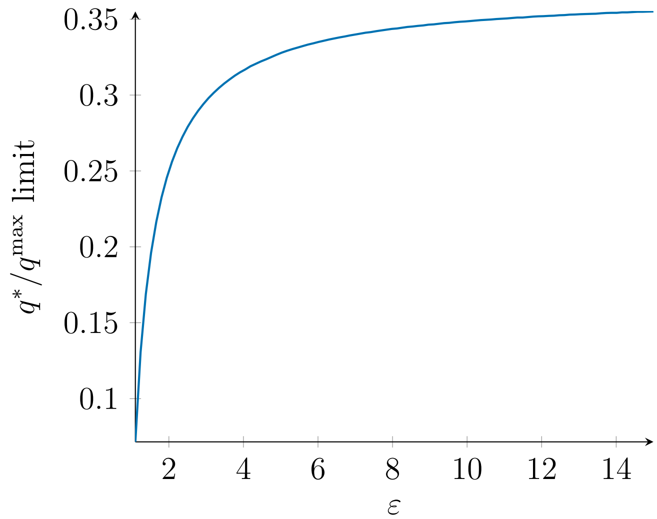


Figure A3: Optimal target block size for a monopolist validator as a function of ε

D Numerical Examples

Methodology: The inference of a demand curve requires random variation in supply to distinguish between shifts along the demand curve and shifts in the demand curve itself. Nevertheless, such random variation in supply is rare because of the programmatically defined rules of blockchains.

Instead, a different strategy, informed by the user demand model presented in Section 3, is adopted. Lemma 1 shows that, for any density f of user valuations, the price elasticity of demand is the tail ratio $\frac{pf(p)}{1-F(p)}$. Specifically, if the distribution is Pareto, the price elasticity is its Pareto tail coefficient. Knowing the Pareto tail of the distribution of user valuations allows the determination of the optimal adjustment parameter from Proposition 4 as the inverse of the Pareto tail coefficient.

In this analysis, transaction gas prices serve as the nearest proxy for user valuations, given the available data. A Pareto distribution is fitted to the empirical distribution of effective transaction gas prices in each randomly selected block to calculate the optimal adjustment rate. The estimation proceeds in two steps. Within each block, the shape parameter is estimated by maximum likelihood weighting each transaction by the gas it uses. The coefficient reported for each subset of blocks is then the average of these block-level estimates; transactions are never pooled across blocks. Estimating block by block and averaging keeps each estimate conditional on the block's demand state, so that block-level demand shifts are not confounded with within-block heterogeneity in valuations—the object required by Proposition 4 is precisely the within-state elasticity at the target. However, this approach is not without limitations.[22] First, gas prices censor user valuations at the lower end of the distribution because of the base fee. Second, pending transactions in the mempool, which carry lower base fees, are not included. Through these two channels, the estimate will reflect a heavier tail than the actual user valuation distribution, biasing the Pareto coefficient down and its inverse—the optimal adjustment rate—up. Third, the effective gas price is mechanically compressed from above: a user pays the minimum of her fee cap and the base fee plus tip, so the valuations of inframarginal users are bunched near the base fee rather than revealed. In principle, the submitted fee cap is the incentive-revealing object and would be the ideal proxy; in practice, fee caps are heavily influenced by wallet defaults that set the cap as a mechanical multiple of the recent base fee, so observed caps mix valuations with buffer rules, and I do not pursue that alternative here. The third limitation biases the tail coefficient in the opposite direction from the first two—upper-end compression thins the measured tail and pushes the implied adjustment rate down—so the net direction of bias is ambiguous a priori, and the point estimates should be treated as a tentative benchmark. Relatedly, within-block dispersion in effective gas prices partly reflects wallet tip-setting heuristics rather than dispersion in user valuations, a further reason to treat the estimates as illustrative of the method rather than as definitive parameters.[23] To address these limitations, several robustness exercises are performed, which include restricting the estimation to blocks with a minimum gas price below a certain threshold (to limit the censoring of low valuations) and to blocks that are less than full (to limit the censoring of mempool transactions). These robustness exercises do not alter the primary conclusions of this numerical analysis.

Discussion: The estimate from a separate 100,000-block sample drawn prior to the merge (see Section 4) suggests an adjustment rate of 6.57% (tail coefficient 15.22), while the corresponding estimate for blocks following the merge is 8.68% (tail coefficient 11.53). The observation that the current adjustment rate of 12.5% overshoots more before the merge

[22]As users demand different amounts of block space, each transaction is weighted by the gas units that it uses to fit the Pareto distribution.

[23]Every estimate across the grid of sample restrictions below lies under the current value of 12.5%, which is the robust content of this analysis for the design of Ethereum's TFM.

than after aligns with the findings of Leonardos et al. (2023). The authors postulate that this discrepancy is because inter-block times were (approximately) exponentially distributed prior to the proof-of-stake upgrade whereas they are constant in Ethereum's proof-of-stake protocol. A constant block arrival rate more accurately reflects our model, suggesting that an estimate around 8% is more suitable for the current blockchain. One might wonder whose price elasticity our estimate represents and why it is so high (over 10). The adjustment rate represents the inverse elasticity of aggregate next-block demand, evaluated at the valuation of the marginal user submitting her transaction for inclusion in the next block. Given the low block size limit, which is set because of technological constraints, it is not implausible that a marginal decentralized finance (DeFi) user with sensitive transactions would exhibit significant demand elasticity, similar to how high-frequency traders are sensitive to spreads.

Another consideration is that this analysis does not account for users whose transactions remain in the mempool for several blocks before confirmation. Notably, under EIP-1559, base fees do not decrease fast enough after a surge in demand. Fee policies that respond to intra-day or intra-hourly demand variations would not need to adjust prices based on the size of the previous block. Instead, it could maintain fixed prices during high-demand periods, fill blocks, and monitor the mempool for price adjustments. However, mempool data are typically not recorded on-chain (i.e., as part of the immutable blockchain) and can be easily manipulated.

Alternative Calculations: This section presents the results of supplementary robustness checks to validate the primary numerical analysis findings. These checks are performed under various conditions to address potential concerns highlighted above, such as the base fee causing censoring of low valuations and the exclusion of pending mempool transactions. To investigate the consistency of the Pareto tail coefficient and the optimal adjustment rate, I adjust the selection of blocks within a size range of $q^{target}(1 \pm \delta\%)$ and different limits on base fees.

The estimations are performed on three samples: the full sample, the pre-merge sample, and the post-merge sample. Tables A2, A3, and A4 provide detailed results. The estimated optimal adjustment rates are consistently lower than the 12.5% adjustment rate and remain stable under various data partitionings. The estimate is smaller for the pre- than for the post-merge sample, as previously found. The adjustment rates decrease as the window around the target block size widens and the maximum base fee increases. Ethereum transaction fees are paid in units of gwei. The increase in rates as the max base fee limit becomes more restrictive (for blocks with a base fee of less than 30 gwei) can only produce thicker tails because of a restricted range and censoring. Variations based on δ, the window of the target

Parameters		Outputs		Observations
δ	max base fee (gwei)	shape α	adjustment rate d	number of blocks
5%	200	12.45	8.03	7189
	100	11.83	8.45	6645
	60	11.01	9.08	5867
	30	9.50	10.53	4130
33%	200	12.77	7.83	43718
	100	12.09	8.27	40471
	60	11.26	8.88	35649
	30	9.48	10.55	25193
87.5%	200	13.64	7.33	78881
	100	12.62	7.92	70769
	60	11.53	8.67	60141
	30	9.50	10.53	41140

Table A2: Shape of Pareto fit α and optimal adjustment rate d for the different maximum gas prices (base fee) in units of gwei and selection of blocks within size $q^{target} \pm \delta\%$ (full sample estimation)

block size, are quite robust, with an adjustment rate in the full sample estimation ranging from 7.33% to 8.03% for a more accommodating max base fee of 200 gwei. Estimates from the pre-merge and post-merge samples suggest that the optimal adjustment rate lies within the 6% to 10% window.

Blockchain designers must adopt simple, robust, and principled methods for updating TFM parameters. However, updating parameters and the "rules of the game" as we go might not be the best track for fostering scalability. Considering nondeterministic adjustment rates for TFMs could provide a solution. Preliminary quantitative explorations using adjustment rates derived from prices and quantities of the preceding two blocks have shown promising results in terms of how well these rates reflect market conditions during the relevant periods. A more stable and manipulation-resistant approach could involve calculating an average elasticity over a range of prior blocks. Alternatively, introducing exogenous noise into the target block size—the supply curve—would generate the supply-side variation needed to trace the local demand curve and estimate its elasticity; with that demand relationship in hand, demand fluctuations could then be recovered.

Parameters		Outputs		Observations
δ	max base fee (gwei)	shape α	adjustment rate d	number of blocks
5%	200	11.46	8.73	5088
	100	11.24	8.89	4962
	60	10.84	9.23	4688
	30	9.55	10.47	3547
33%	200	11.48	8.71	30029
	100	11.29	8.86	29454
	60	10.89	9.18	27884
	30	9.55	10.47	21288
87.5%	200	11.32	8.83	42042
	100	11.15	8.97	41348
	60	10.78	9.28	39426
	30	9.47	10.56	30890

Table A3: Shape of Pareto fit α and optimal adjustment rate d for the different maximum gas prices (base fee) in units of gwei and selection of blocks within size $q^{target} \pm \delta\%$ (post-merge sample estimation)

Parameters		Outputs		Observations
δ	max base fee (gwei)	shape α	adjustment rate d	number of blocks
5%	200	14.85	6.73	2101
	100	13.57	7.37	1683
	60	11.70	8.55	1179
	30	9.15	10.93	583
33%	200	15.59	6.41	13689
	100	14.22	7.03	11017
	60	12.57	7.96	7765
	30	9.09	11.01	3905
87.5%	200	16.29	6.14	36839
	100	14.69	6.81	29421
	60	12.96	7.71	20715
	30	9.58	10.44	10250

Table A4: Shape of Pareto fit α and optimal adjustment rate d for the different maximum gas prices (base fee) in units of gwei and selection of blocks within size $q^{target} \pm \delta\%$ (pre-merge sample estimation)